KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF SCIENCE

DEPARTMENT OF MATHEMATICS

# Hospital Chatbot for Color-Coded Clinical Pathways Using Bidirectional Encoder Representations from Transformers

A PROJECT SUBMITTED TO THE DEPARTMENT OF MATHEMATICS, KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF

BACHELOR OF SCIENCE DEGREE

(MATHEMATICS)

BY

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# DECLARATION

We hereby declare that this submission is our own work towards the award of the Bachelor of Science degree in Mathematics and that, to the best of our knowledge, it contains no material previously published by another person nor material which has been accepted for the award of any other degree of the university, except where due acknowledgment has been made in the text.

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# DEDICATION

To my family, lecturers, and the patients whose care motivates this work.

# ABSTRACT

Effective triage is critical for matching hospital resources to patient acuity. In low- and middle-income countries, including Ghana, hospitals frequently suffer from flow management failures characterized by overcrowding, delayed initial assessments, and acuity discontinuity. To address these operational bottlenecks, this project presents a hospital chatbot powered by a fine-tuned Biomedical Bidirectional Encoder Representations from Transformers (BioBERT) model. The system translates unstructured patient chief complaints into South African Triage Scale (SATS) colours and corresponding clinical pathways.

The underlying mathematical architecture uses tokenisation, continuous vector embeddings, and multi-head self-attention to generate contextualized text summaries, which are passed through a Softmax classification head to predict a categorical distribution over five mutually exclusive Emergency Severity Index (ESI)–style urgency levels. A deterministic mapping function, $f_{\text{SATS}}$, subsequently translates these predictions into actionable triage routing (Red, Orange, Yellow, or Green). To support clinical safety, the pipeline features a mathematical relevance gate to filter non-medical inputs prior to classification. Furthermore, the model acts as a text-first precursor to the standard Triage Early Warning Score (TEWS), supplying an initial pathway card $P(C)$, detailing destination and maximum wait time, before physical vitals can be obtained.

Primary reported metrics are derived from external evaluation on the Medical Information Mart for Intensive Care IV Emergency Department module (MIMIC-IV-ED) and the National Hospital Ambulatory Medical Care Survey (NHAMCS) 2018–22, ensuring evaluation against real-world clinical phrasing rather than synthetic holdouts. On the NHAMCS evaluation set, the baseline BioBERT classifier achieved a five-class accuracy of 47.60% and a 4-colour mapping accuracy of 49.95%, with a median inference latency of approximately 94 ms. To address severe class imbalance and study the safety trade-off between critical-case sensitivity and overall precision, stratified oversampling (text augmentation plus random minority oversampling) was applied during training. This successfully raised Level 1 (Red) recall from 8.00% to 21.71% under external domain shift, though at a corresponding cost to overall accuracy. The system functions strictly as a rapid, nurse-overridable decision-support tool, expediting initial routing while preserving human clinical judgment and objective vital-sign scoring.

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# ABBREVIATIONS

**ASTS** Adult South African Triage Scale

**AVPU** Alert, Voice, Pain, Unresponsive (consciousness scale)

**API** Application Programming Interface

**BERT** Bidirectional Encoder Representations from Transformers

**BioBERT** Biomedical Bidirectional Encoder Representations from Transformers

**CLS** Classification token (`[CLS]`) used as the sequence summary in BERT

**ESI** Emergency Severity Index

**FFN** Feed-Forward Network

**HIS** Hospital Information System

**HIT** Hospital Information Technology

**HR** Heart Rate

**KATH** Komfo Anokye Teaching Hospital

**KNUST** Kwame Nkrumah University of Science and Technology

**LMIC** Low- and Middle-Income Country

**MIMIC** Medical Information Mart for Intensive Care

**MIMIC-IV-ED** MIMIC-IV Emergency Department module (v2.2 schema in this project)

**MLM** Masked Language Modelling

**NHAMCS** National Hospital Ambulatory Medical Care Survey

**NLL** Negative Log-Likelihood

**NLP** Natural Language Processing

**OPD** Outpatient Department

**RR** Respiratory Rate

**SATS** South African Triage Scale

**SBP** Systolic Blood Pressure

**TEWS** Triage Early Warning Score

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# CHAPTER 1

# INTRODUCTION

## 1.1 Background

Hospitals in Ghana operate under sustained pressure: crowded waiting areas, limited nursing staff, and long queues before a clinician sees a new arrival. Through triage, each arriving patient is assigned an urgency level so that scarce beds, nurses, and clinicians match clinical need rather than queue order alone. In well-resourced systems, urgency is reassessed as patients move through diagnostics, wards, and outpatient streams. In resource-constrained settings, however, standardised protocols such as the South African Triage Scale (SATS) are often applied only at the first desk. Once a patient leaves that point, their colour-coded acuity may no longer steer bed allocation, laboratory orders, or re-assessment timers. We call this failure *acuity discontinuity*.

Hospitals in Ghana, including KNUST University Hospital, apply SATS to structure intake across clinical departments, not only in a single emergency bay. Colour codes (Red, Orange, Yellow, Green, and Blue for dignity protocols) express how quickly a patient must be seen and where they should stream next: resuscitation, acute beds, urgent waiting, or routine outpatient care. Nurses at triage also compute a Triage Early Warning Score (TEWS) from objective vital signs. Yet many delays begin earlier, when a patient arrives with only words: a headache, fever, weakness, or abdominal pain described in free text before any thermometer or pulse oximeter reading is available.

This project proposes a digital *Flow Management Engine* in the form of a hospital chatbot for color-coded clinical pathways using Bidirectional Encoder Representations from Transformers (BERT). The chatbot accepts a single English free-text *chief complaint* (the primary reason the patient presents, in their own words; see Section 3.1.1) and returns a SATS colour reflecting urgency together with acuity probabilities and a clinical pathway card that can guide routing across outpatient departments, wards, and intake queues. The core contribution is mathematical: a pipeline from unstructured complaint text to acuity probabilities, colour, and pathway card, developed in Chapter 3. The deployed encoder is BioBERT, a biomedical BERT checkpoint fine-tuned for five-class triage. Crucially, this system is engineered strictly as a rapid, nurse-overridable decision-support tool. It expedites initial routing and queue management while strictly preserving the primacy of human clinical judgment and objective vital-sign scoring.

## 1.2 Motivation and significance

Three operational facts motivate the work.

1. **Text arrives before vitals.** At home, in a queue, or at a kiosk, patients can describe symptoms long before six TEWS parameters are measured. A language model that produces an initial SATS colour therefore fills a real temporal gap between arrival and nurse assessment (Stewart et al., 2023).

2. **Colour must travel with the patient.** SATS colours already exist in Ghanaian hospital policy (Rominski et al., 2014; KATH Emergency Department, 2018). The unsolved problem is not inventing colours, but binding them to destination, maximum waiting time $T_{\max}$ (the longest time allowed for that colour before the patient should be seen or reassessed), and escalation so that urgency does not vanish after the intake desk (South African Triage Group, 2012; Sundström et al., 2014).

3. **Green congestion harms Red and Orange care.** Non-urgent presentations occupying acute capacity (sometimes called the Green Tag Burden) delay time-critical patients (Yilmaz, 2021). Digital diversion of Green pathways toward outpatient or specialty streams is therefore a flow-management requirement, not a convenience feature.

For mathematics students and hospital stakeholders alike, the significance is twofold. First, the project shows how contemporary Transformer mathematics (tokenisation, embeddings, self-attention, Softmax) can be stated rigorously enough for examination while remaining deployable as a chatbot API. Second, it links that mathematics to an auditable protocol for healthcare facilities: every predicted acuity level maps through a fixed function $f_{\text{SATS}}$ (which converts acuity levels 1–5 into a SATS colour) and then to a pathway card $P(C)$ (destination, $T_{\max}$, actions, and escalation for that colour).

## 1.3 Problem statement

Low- and middle-income country (LMIC) hospitals struggle with crowded waiting rooms and limited human resources at intake. Patients may wait hours before first assessment. Critical cases are not always identified quickly, while less urgent presentations congest acute zones. Medical systems are confusing; without immediate direction, patients do not know whether to join the emergency stream, the polyclinic, or self-care advice.

Manual, paper-based triage creates acuity discontinuity, rendering patient status static rather than dynamically responsive to clinical deterioration. This discontinuity manifests in five systemic failures:

1. **Pre-Assessment Bottleneck.** Critical delays accumulate prior to any clinical evaluation because the initial human assessor is fundamentally capacity-constrained.

2. **Acuity Degradation.** Patients correctly triaged as Orange or Red frequently lose their time-critical priority when transitioning to imaging, laboratories, or wards, as paper records fail to enforce dynamic countdown timers across departments.

3. **Low-Acuity Congestion.** Non-urgent (Green) presentations lack automated diversion toward outpatient or self-care routes, inadvertently occupying acute beds and human resources required for life-saving interventions.

4. **Vital-Sign Latency.** At arrival, objective measurements are rarely complete. Subjective, high-risk language (e.g. "worst headache of my life") remains computationally unutilized by TEWS until physical vitals are subsequently recorded.

5. **Protocol Misalignment.** Existing generic symptom checkers fail to map predictions to standardized SATS hospital pathways or local operational routing, rendering their outputs unactionable at the facility level.

Without a framework to manage flow hospital-wide, overcrowding and resource mismatch persist; the most vulnerable patients become invisible to the system after the first desk.

## 1.4 Aim and objectives

### 1.4.1 General objective

To design and specify a hospital chatbot that maps an unstructured English chief complaint to a SATS colour-coded clinical pathway, using a fine-tuned BioBERT classifier and protocol tables aligned to Ghanaian hospital intake.

### 1.4.2 Specific objectives

1. **Build a BioBERT classifier.** Predict an ESI-style urgency level (1–5) from a single unstructured English chief complaint using a fine-tuned BioBERT Softmax classification head.

2. **Deploy colour and pathway at intake.** Return a SATS colour and a pathway card from text alone before vitals are taken, with TEWS kept as the parallel nurse vital-sign standard.

3. **Add a clinical gate.** Reject non-clinical chat so that non-medical messages receive no colour before BioBERT runs.

## 1.5 Research questions

Transitioning the theoretical capabilities of Transformer language models into a practical clinical tool requires systematic validation. To evaluate the computational accuracy, operational utility, and clinical safety of the proposed Flow Management Engine, this research seeks to answer the following three questions.

1. Can a fine-tuned BioBERT model accurately classify unstructured English chief complaints into the correct ESI-style urgency level (1–5) under external domain shift?

2. Can a deterministic mapping heuristic ($f_{\text{SATS}}$) successfully provision SATS colour and pathway guidance from text prior to the acquisition of physical vitals?

3. Can a mathematical clinical relevance gate reliably detect and reject non-medical inputs before executing the Transformer classification model?

# CHAPTER 2

# LITERATURE REVIEW

## 2.1 Introduction

A comprehensive evaluation of published literature provides the necessary context for this project, encompassing hospital overcrowding, SATS and TEWS protocols, clinical pathways, Transformer language models, and natural language processing (NLP) in triage settings. Organized thematically, the subsequent review brings together current knowledge by allowing each section to build logically upon the last. This structure is deliberately designed to highlight the practical implications for deploying a triage chatbot within the Ghanaian healthcare system. Concluding the review, an explicit research gap is identified, establishing the theoretical justification for the text-based computational pipeline addressed in Chapter 3.

## 2.2 Hospital overcrowding and triage delay

Overcrowding in low- and middle-income country (LMIC) hospitals affects outpatient departments, emergency intake, and inpatient wards simultaneously (KATH Emergency Department, 2018; Mitchell et al., 2023). Delayed triage lengthens time-to-provider, increases left-without-being-seen rates, and correlates with poor outcomes for time-sensitive conditions. Waiting time interacts with colour assignment: when reassessment is weak, deterioration after the first desk can go unnoticed (Nurchis et al., 2025).

A structural driver is competition for acute capacity by non-urgent patients: the Green Tag Burden (Yilmaz, 2021). In LMICs, understaffing and limited diagnostics intensify that competition. Digital diversion of Green patients to outpatient, minors, or specialty streams is therefore a flow-management requirement for colour-coded pathways, not merely a convenience feature (Sundström et al., 2014).

**Implication for this project.** Any chatbot that assigns SATS colours must also attach destinations and timers; otherwise colour remains a label that dies at the desk (acuity discontinuity in Chapter 1). The next section reviews how SATS already supplies those colours and waiting-time rules in Ghanaian practice.

## 2.3 SATS in LMIC and Ghanaian practice

The South African Triage Scale (SATS) provides a five-colour framework (Red, Orange, Yellow, Green, Blue) with maximum waiting times $T_{\max}$ (the longest time allowed for each colour before the patient should be seen or reassessed) and destination rules (South African Triage Group, 2012). Implementation at Komfo Anokye Teaching Hospital (KATH) and related Ghanaian settings demonstrated that standardised colour coding can structure flow even when staffing is constrained, provided urgency information travels with the patient (Rominski et al., 2014; KATH Emergency Department, 2018).

SATS combines physiological early warning (TEWS) with clinical discriminators that can escalate colour when high-risk symptoms are present even if vitals are only mildly abnormal (for example central chest pain). Colour codes therefore already exist in hospital policy. The research gap is not inventing colours, but (i) assigning them from free-text complaints before full vitals, and (ii) binding colours to pathway protocols so acuity does not disappear after intake.

Hospitals such as KNUST University Hospital apply similar colour logic across clinical departments, not only in a single emergency bay. That multi-department setting motivates pathway cards that name outpatient urgent streams, surgery, obstetrics, pediatrics, and support units rather than a single undifferentiated "ED" destination.

Colour alone still needs an objective vital-sign companion when measurements exist. The next section therefore turns to TEWS, the nurse-facing score that runs in parallel with any text-based colour.

## 2.4 TEWS and vital-sign triage

The Triage Early Warning Score (TEWS) is the deterministic vital-sign score used at SATS desks (South African Triage Group, 2012). For adult patients, nurses record seven parameters: mobility, respiratory rate (RR), heart rate (HR), systolic blood pressure (SBP), temperature, Alert–Voice–Pain–Unresponsive (AVPU) consciousness, and a trauma flag. Each parameter is converted into a small integer of points using the SATS lookup tables (for example, very high heart rate, hypotension, or unresponsive consciousness contributes more points than a normal reading). Those seven contributions are then added to form a single total score $T$.

That integer $T$ maps to a colour band: Red for $T \geq 7$, Orange for $5 \leq T \leq 6$, Yellow for $3 \leq T \leq 4$, and Green for $T \leq 2$. Because the map is fixed and auditable, two nurses who measure the same vitals obtain the same $T$ and the same colour. TEWS is therefore objective; it does not read language. The formal sum and piecewise maps $f_k$ are derived in Section 3.3.

Subjective symptoms such as "worst headache of my life" or "feverish and weak" may appear in the complaint long before corresponding vitals exist. TEWS cannot score a patient who has not yet been measured. Chapter 3 therefore presents TEWS mathematics as the hospital standard *parallel* to the text model: when vitals are available, nurses compute $T$; when they are not, the BioBERT path supplies an initial colour for routing assist. This project does not claim that Softmax replaces TEWS.

Colour and TEWS still need an operational destination. The next section reviews clinical pathways as the roadmaps that turn colour into where the patient should go and how soon.

## 2.5 Clinical pathways as operational roadmaps

Clinical pathways are step-by-step care roadmaps: destination, diagnostics, time targets, and escalation (Sundström et al., 2014). In SATS terms, colour without a pathway is incomplete: Red must mean immediate resuscitation access; Orange must mean a short countdown; Green must mean diversion away from acute beds when clinically appropriate (South African Triage Group, 2012; Yilmaz, 2021).

Pathway literature supports treating colour as operational, not decorative. Under-triage (assigning too low an urgency) is a recognised quality trigger in emergency medicine (Berkowitz et al., 2019); over-triage wastes scarce acute capacity. A chatbot that emits pathway cards $P(C)$ (for colour $C$: destination, $T_{\max}$, actions, and escalation) must therefore be evaluated not only on label accuracy but on whether routing rules are protocol-faithful and nurse-overridable.

To assign colour and pathway from free text before vitals, the project needs a language model that understands clinical wording. The next section reviews the Transformer mathematics that makes that possible.

## 2.6 Attention, BERT, and BioBERT

Sequential recurrent language models read text predominantly in one direction, limiting context for ambiguous clinical phrases. Vaswani et al. (2017) introduced scaled dot-product self-attention, allowing each token to attend to all others in parallel within a fixed window. That property matters for triage language, where negation and severity modifiers (“no pain”, “severe headache”) must reshape neighbouring words.

Devlin et al. (2019) built BERT (Bidirectional Encoder Representations from Transformers) on this idea: masked language modelling (MLM) pre-training yields contextual embeddings

$$E(t_i) = E_{\text{word}}(t_i) + E_{\text{pos}}(i) + E_{\text{seg}}(s)$$

where $t_i$ is token $i$, $E_{\text{word}}$, $E_{\text{pos}}$, and $E_{\text{seg}}$ are word, position, and segment embeddings, and $s$ is the segment id (defined fully in Chapter 3). BERT is an encoder stack: it produces contextual vectors for classification and tagging rather than open-ended generation.

Lee et al. (2020) continued pre-training on PubMed and PubMed Central text to produce Biomedical BERT (BioBERT), a domain-specific encoder whose embedding space places clinical terms nearer related concepts. Fine-tuning BioBERT on nurse-labeled chief complaints yields a supervised classifier from text to acuity levels. Classification with a fixed five-class label set reduces open-ended generative hallucination risk compared with unconstrained large language models that invent diagnoses in free prose (Stewart et al., 2023). The fixed map $f_{\text{SATS}}$ then converts the predicted acuity level into a SATS colour for pathway routing.

Those models must still be judged against published triage NLP evidence. The next section reviews what that literature already shows and where it stops short for Ghanaian intake.

## 2.7 NLP for emergency and hospital triage

Recent evaluations of natural language processing in emergency triage highlight chief-complaint classification and clinical decision support as highly active, yet unevenly validated, research domains. For instance, Gligorijević et al. (2018) demonstrated the efficacy of deep attention models in stratifying emergency department patients based purely on unstructured text.

However, a critical vulnerability across these diagnostic models is severe class imbalance. Rare, life-threatening presentations (Level 1 / Red) are chronically under-represented compared to routine Yellow and Green cases. As a result, optimizing these models for global accuracy alone can obscure clinically unsafe under-triage, necessitating specialized statistical strategies to reduce the imbalance (Berkowitz et al., 2019; Chawla et al., 2002).

Furthermore, the majority of published NLP architectures are trained on high-income clinical corpora or electronic health records authored by medical professionals, rather than raw patient phrasing. Most open-access triage benchmarks, such as MIMIC-IV-ED (Johnson et al., 2023a) and NHAMCS (National Center for Health Statistics (NCHS), 2024), are labelled using the five-level Emergency Severity Index (ESI) predominantly used in North America. The literature lacks a formalized framework to computationally bridge these resource-based ESI predictions to the physiological, colour-coded SATS pathways required for Ghanaian hospital intake, and it likewise lacks a mathematical pipeline that maps unstructured patient free text directly to SATS colours while binding those predictions to local departmental routing structures. While synthetic benchmarks such as Triagegeist (Laitinen-Fredriksson Foundation, 2025) and US-based datasets enable large-scale supervised fine-tuning, they introduce geographic and demographic domain shift. They remain a foundational computational step and do not circumvent the need for rigorous, prospective local validation in Ghanaian facilities.

Accuracy on labels is not the whole product. The next section reviews conversational triage tools and why this project's chatbot is deliberately constrained to colour and pathway cards.

## 2.8 Conversational agents and digital triage tools

Symptom-assessment applications and ED prediction chatbots show that conversational NLP can support triage-like advice before the nurse desk (Stewart et al., 2023). Validation studies emphasise safety, under-triage monitoring, nurse override, and comparison with conventional triage. Consumer tools often return free-form advice that is not mapped to hospital colour protocols or local destinations.

Our chatbot is deliberately constrained: it outputs acuity probabilities, a SATS colour, and a pathway card $P(C)$ aligned to SATS and Ghanaian facility routing, not free-form diagnosis essays. A clinical relevance gate rejects non-medical chat so that BioBERT is not run on greetings or sports conversation. That constrained design follows the safety emphasis in the triage NLP review literature.

## 2.9 Summary: what prior work supplies and what it leaves open

Table 2.1 summarises the strands reviewed above and shows how each strand feeds the next.

| Strand | What exists | Open for this project |
|---|---|---|
| Overcrowding / Green burden | Evidence that delay and non-urgent congestion harm flow (Mitchell et al., 2023; Yilmaz, 2021) | Digital colour + diversion rules at Ghanaian intake |
| SATS / TEWS | Colour bands and vital-score algebra (South African Triage Group, 2012; Rominski et al., 2014) | Text-first colour before vitals; TEWS kept parallel |
| Pathways | Operational roadmaps and diversion models (Sundström et al., 2014) | Explicit $P(C)$ tables for seventeen departments |
| Transformers / BioBERT | Contextual encoders for biomedical text (Vaswani et al., 2017; Devlin et al., 2019; Lee et al., 2020) | Transparent pipeline $X \rightarrow \hat{y} \rightarrow C \rightarrow P(C)$ |
| Triage NLP | ED text classifiers and reviews (Stewart et al., 2023; Gligorijević et al., 2018) | SATS-aligned, pathway-bound chatbot for LMIC intake |

Table 2.1: Literature summary: available foundations versus the gap this project fills.

## 2.10 Research gap

Existing work separates vital-sign scores (TEWS), manual SATS colour assignment, and NLP classifiers trained largely abroad. There is no unified mathematical account of how a free-text English chief complaint at Ghanaian hospital intake becomes a SATS colour and pathway card using BioBERT, while TEWS remains documented as the nurse-facing vital standard and nurses retain override.

In short, prior art provides colours, scores, and growing NLP evidence; it does not provide the integrated text path

$$X \xrightarrow{\tau} H^{(0)} \xrightarrow{\text{BioBERT}} \hat{\mathbf{y}} \xrightarrow{f_{\text{SATS}}} C_{\text{NLP}} \rightarrow P(C_{\text{NLP}})$$

where $X$ is the chief complaint, $\tau$ is tokenisation, $H^{(0)}$ is the input embedding matrix, $\hat{\mathbf{y}}$ is the acuity distribution, $f_{\text{SATS}}$ is the fixed map from acuity level to SATS colour $C_{\text{NLP}}$, and $P(C_{\text{NLP}})$ is the pathway card (destination, $T_{\max}$, actions, escalation), all developed in Chapter 3, with clinical gating, Softmax framed as a Categorical distribution over mutually exclusive acuity levels, and department routing tables matched to Ghanaian healthcare facilities.

This project addresses three specific gaps in the current triage literature:

1. **Transparent Mathematics.** Many NLP triage papers report final accuracy without showing the underlying mathematical steps. This project explicitly details every transformation, from tokenisation to the classification Softmax, in a consistent notation.

2. **Actionable Pathways.** Most models output a bare urgency label and stop. This project binds that label to a SATS-aligned pathway, providing specific waiting times, escalation rules, and department destinations mapped to a Ghanaian hospital setting.

3. **Parallel TEWS Integration.** Some literature treats text classifiers as a complete replacement for vital signs. This project explicitly positions the Transformer text-path as an antecedent routing assist, retaining TEWS as the independent, deterministic standard for objective vital-sign scoring.

Chapter 3 fills that gap for the text path; Chapter 4 reports evaluation.

# CHAPTER 3

# METHODOLOGY

Translating unstructured chief complaints into actionable South African Triage Scale (SATS) classifications requires a rigorous computational pipeline. To bridge the underlying mathematical architecture with its practical application, the following sections employ a recurring illustrative example modeled on a standard patient presentation at KNUST University Hospital: *"I have a headache and feel feverish. My body aches and I feel weak."* This end-to-end transformation applies to all textual inputs that successfully clear the clinical relevance gate detailed in Section 3.17. Crucially, this architecture operates strictly as a clinical decision-support mechanism. It provides probabilistic routing guidance to assist triage staff, rather than functioning as an autonomous diagnostic tool or replacing definitive clinical judgment.

## 3.1 Methodological overview

The hospital chatbot draws on two complementary ideas from clinical practice; this project develops the text-based path in full and documents the Triage Early Warning Score (TEWS) as the parallel vital-sign standard.
The chapter addresses the three specific objectives from Chapter 1 in order:

1. Build a BioBERT Softmax classifier that turns one English chief complaint into urgency probabilities over levels 1–5. The chapter specifies tokenisation, embeddings, twelve encoder layers with self-attention, the classification head, and Stage 3 fine-tuning so the mathematics from text to acuity is transparent end to end.

2. Deploy colour and pathway guidance from text at intake before vitals are taken. After the Softmax chooses a level, the fixed map $f_{\text{SATS}}$ names the SATS colour and the pathway card $P(C)$ states destination, maximum waiting time $T_{\max}$, actions, and escalation, while TEWS remains the nurse vital-sign path in parallel.

3. Add a clinical relevance gate so non-clinical chat receives no colour. Before BioBERT runs, the gate scores how clinical the message is and withholds colour from greetings and other non-medical text.

First, **clinical pathways** encode hospital protocols: symptoms and context map to an urgency colour and a destination (resuscitation bay, acute bed, urgent waiting, outpatient stream). Pathways are the product the chatbot shows to patients and staff after classification.

Second, TEWS is the nurse-facing vital-sign score used at SATS triage desks. When heart rate (HR), respiratory rate (RR), systolic blood pressure (SBP), mobility, temperature, Alert, Voice, Pain, Unresponsive (AVPU) consciousness, and trauma status are measured, TEWS produces a deterministic integer $T$ and colour band $C_{\text{TEWS}}(T)$. TEWS is explained fully in Section 3.3 because it is the objective hospital standard; the chatbot's primary engine for *free-text chief complaints* at intake is a deep attention model (Biomedical BERT, BioBERT), not a weighted sum of vitals.

When vitals are not yet available, the text model below supplies the intake colour. Nurses may later record vitals and apply TEWS independently; this project specifies the natural language processing (NLP) path only.

### 3.1.1 Chief complaints

The *chief complaint* is the primary reason the patient presents, expressed in their own words (or an attendant's) as unstructured free text at intake, before vitals, physical examination, or formal diagnosis. Nurses and triage staff traditionally record this as prose on paper or electronic forms; it captures symptom onset, severity, and context (for example, "headache and feverish, body aches, feel weak"). It is not a diagnosis, not a TEWS vital-sign score, and not a full medical history: it is only the presenting problem text $X$ that this project classifies.

The chatbot accepts one **English** chief-complaint string per submission. The running example *"I have a headache and feel feverish. My body aches and I feel weak."* illustrates a typical Yellow-level presentation used throughout this chapter; pipeline symbol $X$ appears in Table 3.1 and Figure 3.1.

### 3.1.2 Training data provenance

Model training and external evaluation draw on distinct data layers.

**Training corpus (not published test).** Stage 3 fine-tuning uses the Triagegeist synthetic emergency-department benchmark released by the Laitinen-Fredriksson Foundation on Kaggle (Laitinen-Fredriksson Foundation, 2025). Structured intake records (`train.csv`, acuity labels) are inner-joined with free-text rows (`chief_complaints.csv`) on `patient_id`, yielding $N = 80{,}000$ pairs of English free-text chief complaints with nurse-assigned ESI-style urgency levels 1–5. Vitals and demographics in the source files are excluded so the NLP path matches single-shot text input at intake (Section 3.14.3). Labels map to SATS colours only via the proposed heuristic $f_{\text{SATS}}$ (Section 3.15); the corpus is **not** SATS-labelled. This corpus was subjected to a stratified 90/10 split (seed 42), resulting in approximately 72,000 pairs for training and 8,000 pairs for internal validation. The 72,000-pair training set was used exclusively to update the model weights during backpropagation. The 8,000-pair validation set was isolated strictly to monitor validation loss and select the optimal training checkpoint. Crucially, neither of these Triagegeist subsets was used to evaluate the model's final clinical performance. To ensure a rigorous evaluation completely devoid of data leakage or synthetic memorization, all published accuracy, colour accuracy, and under/over-triage metrics in Chapter 4 are evaluated entirely on separate, external real-world cohorts: the MIMIC-IV-ED open demo matching the v2.2 triage schema (Johnson et al., 2023b,a) and NHAMCS 2018–22 (National Center for Health Statistics (NCHS), 2024). This corpus was **not** collected at KNUST University Hospital.

**KNUST-informed contextual examples.** Following an observational site visit to KNUST University Hospital, the authors mapped existing clinical pathways and department structures through direct observation of intake workflows and stakeholder inquiries with hospital staff. Based on these established protocols, the authors generated supplementary English chief complaints to validate pathway routing and illustrate the pipeline (for example, *"I have a headache and feel feverish. My body aches and I feel weak."* in Table 3.22); this supports qualitative analysis only. They are **not** merged into the 80,000-pair fine-tuning corpus.

### 3.1.3 BERT and bidirectional context

Older language models read text sequentially (left-to-right or right-to-left). Vaswani et al. (2017) showed that *self-attention* lets every token attend to every other token in one pass. Devlin et al. (2019) built Bidirectional Encoder Representations from Transformers (BERT) on this mechanism: during pre-training, random tokens are masked and predicted from full left *and* right context via masked language modelling (MLM). The result is contextual embeddings: the vector for "feverish" differs depending on neighbouring words such as "headache" or "not".

### 3.1.4 BioBERT for biomedical text

General BERT embeddings under-represent clinical terms. Lee et al. (2020) continued MLM pre-training on PubMed and PMC, then released BioBERT with the same transformer skeleton as BERT but an embedding space tuned for biomedical language. For triage, BioBERT is fine-tuned on the Triagegeist corpus described in Section 3.1.2: supervised learning maps English complaint text $X$ to acuity level probabilities $\hat{\mathbf{y}}$, then to SATS colour via a fixed map $f_{\text{SATS}}$.

### 3.1.5 The [CLS] summary token

Classification does not read all token outputs separately. An artificial token [CLS] is inserted at position $i = 0$. After $L = 12$ encoder layers, its hidden state $h_{[\mathrm{CLS}]} \in \mathbb{R}^{768}$ summarises the entire complaint and feeds the softmax classification head (Section 3.8). The [CLS] row accumulates context from every other token through the stacked encoder layers.

### 3.1.6 Pipeline summary

Figure 3.1 previews the end-to-end natural language processing (NLP) path from a free-text chief complaint to a SATS colour and pathway card. Each box corresponds to one transformation; the subsections and sections that follow derive the formulas, symbols, and worked steps for every stage.

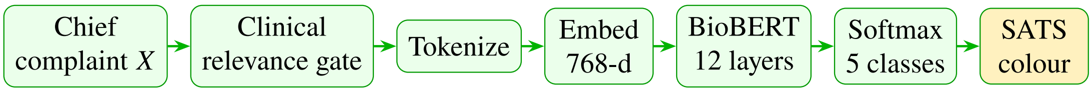


Figure 3.1: NLP pipeline from chief complaint to SATS colour (TEWS is documented separately in Section 3.3).

1. **Chief complaint $X$.** The patient or attendant submits one unstructured English text string describing symptoms and relevant history. That string is the only clinical input to the NLP path at this stage: no vital signs, no diagnosis, and no full medical history are required before the model can propose a colour.

2. **Clinical relevance gate.** A rule-based score $g(X)$ decides whether the message is clinical enough to classify. Greetings and other non-medical chat are rejected here so BioBERT does not invent an urgency colour for text that is not a chief complaint.

3. **Tokenize.** WordPiece splits $X$ into subword tokens and inserts the special markers [CLS], [SEP], and <PAD>. The result is a fixed-length sequence of integer IDs that the embedding table can look up.

4. **Embed (768-d).** Each token ID becomes a 768-dimensional vector. Positional and segment embeddings are added element-wise so the model knows both what each token means and where it sits in the complaint, forming the input matrix $H^{(0)}$.

5. **BioBERT (12 layers).** Twelve encoder layers apply multi-head self-attention and feed-forward transforms. Each layer lets every token look at the others, so symptom words and severity modifiers reshape one another before classification.

6. **Softmax (5 classes).** The `[CLS]` summary vector $h_{[\mathrm{CLS}]}$ passes through a linear head that produces five logits. Softmax turns those logits into acuity probabilities $\hat{\mathbf{y}}$ that add to one; the largest probability is the predicted urgency level.

7. **SATS colour and pathway.** The predicted level $\hat{c}$ maps to colour $C_{\mathrm{NLP}} = f_{\mathrm{SATS}}(\hat{c})$ through the fixed acuity-to-colour map, and the pathway card $P(C_{\mathrm{NLP}})$ names destination, $T_{\max}$, actions, and escalation for that colour.

Each stage above is derived and exemplified later in this chapter. The end-to-end summary restates the full path for the implemented prototype.

### 3.1.7 Safety and nurse authority

The Flow Management Engine is engineered strictly as a clinical decision-support system; it is structurally constrained from acting as an autonomous diagnostic or discharge authority. To preserve the primacy of human clinical judgment, the architecture enforces three strict operational boundaries throughout the pipeline:

1. **Algorithmic Gating.** A mathematical relevance gate actively filters non-clinical text; non-medical inputs are systematically rejected prior to BioBERT classification.

2. **Probabilistic Output Constraint.** The classification head outputs a Categorical distribution $\hat{\mathbf{y}}$ rather than a binary safety certificate. Low-confidence classifications and near-ties explicitly mandate immediate escalation to human triage staff.

3. **Preservation of Objective Scoring.** When physical vitals are acquired, TEWS and SATS discriminators govern bedside clinical colour. The NLP text-path operates as an antecedent routing assist and is programmed to never overwrite objective physiological data.

Pathway cards therefore propose destinations and timers that nurses can accept, modify, or reject. That principle is repeated in the results discussion (Section 4.5) and when Softmax confidence is low.
Table 3.1 records the mathematical object entering and leaving each stage so the reader can follow how one complaint string becomes a colour and pathway card.

| Stage | Input | Output |
|---|---|---|
| Chief complaint | Patient text string | $X$ (raw string) |
| Clinical gate | $X$ | Pass/fail via $g(X)$ |
| Tokenize | $X$ | Token sequence $(t_1, \ldots, t_M)$, $M \le 128$ |
| Embed | Token IDs | Input matrix $H^{(0)} \in \mathbb{R}^{M \times 768}$ |
| BioBERT | $H^{(0)}$ | Final matrix $H^{(12)}$; summary $h_{[\text{CLS}]}$ |
| Softmax | $h_{[\text{CLS}]}$ | Acuity probabilities $\hat{\mathbf{y}} \in \mathbb{R}^5$ |
| SATS colour | $\hat{\mathbf{y}}$ | Level $\hat{c}$, colour $C_{\text{NLP}}$, pathway $P(C_{\text{NLP}})$ |

Table 3.1: Pipeline stage inputs and outputs (NLP path).

## 3.2 SATS colour chart

SATS assigns each living patient an urgency colour with a maximum waiting time $T_{\max}$ and hospital destination (South African Triage Group, 2012). Figure 3.2 separates immediate and special protocols from the decreasing-urgency scale used for living patients. Blue indicates deceased-on-arrival dignity protocol, not low urgency.

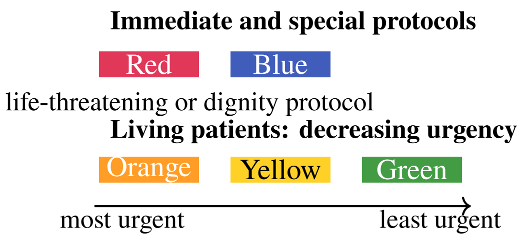


Figure 3.2: South African Triage Scale (SATS) colour bands used at hospital intake.

| Colour | Meaning | $T_{\max}$ | Typical destination |
|---|---|---|---|
| Red | Immediate life threat | 0 min | Resuscitation bay |
| Orange | Very urgent | 10 min | Acute / high-dependency bed |
| Yellow | Urgent | 60 min | Emergency waiting / urgent stream |
| Green | Non-urgent | $< 4$ h | Outpatient department (OPD) / minors / polyclinic |
| Blue | Dignity protocol | N/A | Mortuary / private space |

Table 3.2: SATS colours, waiting targets, and routing (SATS-aligned).

**Remark 3.1.** *Red signals an immediate threat to life; Green indicates that the patient* *may safely wait in a routine stream. The chatbot outputs one of these colours so intake staff can route patients without re-interpreting free text.*

For the illustrative complaint in this chapter, the NLP model in Section 3.13 may assign moderate urgency (Yellow) with high probability; the same framework applies regardless of presenting symptoms.

## 3.3 TEWS: Triage Early Warning Score

TEWS is the deterministic score nurses compute from seven vital parameters at adult SATS (ASTS) triage (South African Triage Group, 2012). It provides an objective colour band when measurements exist. This project documents the adult score; paediatric TEWS tables differ and are out of scope without an age feature.

### 3.3.1 Definition

Let $\mathbf{v} = (v_1, \ldots, v_7)$ denote heart rate (HR), respiratory rate (RR), systolic blood pressure (SBP), mobility, temperature, AVPU consciousness, and trauma flag. Each parameter maps to points $f_k(v_k) \in \{0, 1, 2, 3\}$ via SATS lookup tables. TEWS aggregates these contributions into one integer so that nurses at different desks can compare patients on a common scale. Weights $w_k = 1$:

$$T = \sum_{k=1}^{7} w_k \, f_k(v_k) = \sum_{k=1}^{7} f_k(v_k). \tag{3.1}$$

Summing seven bounded terms yields a score $T \in [0, 21]$ in principle, though typical values at triage are much lower. Because each $f_k$ is defined from the SATS manual, the score is reproducible and auditable: two nurses who measure the same vitals obtain the same $T$.

| $k$ | Parameter $v_k$ | Source | Unit / scale |
|---|---|---|---|
| 1 | Heart rate (HR) | SATS manual, TEWS table | beats/min |
| 2 | Respiratory rate (RR) | SATS manual, TEWS table | breaths/min |
| 3 | Systolic blood pressure (SBP) | SATS manual, TEWS table | mmHg |
| 4 | Mobility | SATS manual | walking / assisted / immobile |
| 5 | Temperature | SATS manual | °C |
| 6 | AVPU consciousness | SATS manual | alert / verbal / pain / unresponsive |
| 7 | Trauma | SATS manual | present / absent |

Table 3.3: TEWS parameters and documentation source (adult SATS).

**Remark 3.2.** *Each vital sign contributes zero to three points. Elevated HR, abnormal RR, low or very high SBP, or reduced consciousness increase $T$; normal values contribute zero.*

### 3.3.2 Piecewise scoring functions

The piecewise functions $f_1$, $f_2$, and $f_3$ encode tachycardia, bradycardia, abnormal respiratory rate, and abnormal systolic pressure as higher point values, reflecting physiological stress seen at triage.

Heart rate $f_1$ (beats/min), where $v_1$ is the measured heart rate:

$$f_1(v_1) = \begin{cases} 3 & v_1 \geq 130 \\ 2 & 111 \leq v_1 \leq 129 \text{ or } v_1 \leq 40 \\ 1 & 101 \leq v_1 \leq 110 \text{ or } 41 \leq v_1 \leq 50 \\ 0 & 51 \leq v_1 \leq 100 \end{cases} \tag{3.2}$$

Elevated heart rate (tachycardia) or very low heart rate therefore increases $T$ and may raise the TEWS colour band.

Respiratory rate $f_2$ (breaths/min), where $v_2$ is the measured respiratory rate:

$$f_2(v_2) = \begin{cases} 3 & v_2 \geq 30 \text{ or } v_2 \leq 8 \\ 2 & 21 \leq v_2 \leq 29 \\ 1 & 9 \leq v_2 \leq 14 \\ 0 & 15 \leq v_2 \leq 20 \end{cases} \tag{3.3}$$

Very high or very low respiratory rate adds points, capturing respiratory distress before full clinical assessment.

Systolic blood pressure $f_3$ (mmHg), where $v_3$ is the measured SBP (South African Triage Group, 2012):

$$f_3(v_3) = \begin{cases} 2 & v_3 > 170 \\ 0 & 101 \leq v_3 \leq 170 \\ 1 & 90 \leq v_3 \leq 100 \\ 3 & v_3 < 90 \end{cases} \tag{3.4}$$

Hypotension ($v_3 < 90$) contributes three points; marked hypertension ($v_3 > 170$) contributes two. The adult normal band 101–170 mmHg contributes zero.

Mobility $f_4$: walking = 0; assisted = 1; immobile = 2–3 (SATS manual). Temperature $f_5$: 35.0–38.4 °C = 0; borderline fever or hypothermia = 1; extreme values = 2–3. AVPU $f_6$: alert = 0, verbal = 1, pain = 2, unresponsive = 3. Trauma $f_7$: none = 0;

major trauma = 2–3.

### 3.3.3 Colour map

The mapping $C_{\mathrm{TEWS}}(T)$ converts the integer score into the same colour vocabulary used at the SATS desk, so vital-sign triage aligns with hospital routing rules (South African Triage Group, 2012):

$$C_{\mathrm{TEWS}}(T) = \begin{cases} \text{Red} & T \geq 7 \\ \text{Orange} & 5 \leq T \leq 6 \\ \text{Yellow} & 3 \leq T \leq 4 \\ \text{Green} & T \leq 2 \end{cases} \tag{3.5}$$

where $T$ is the TEWS total from Equation (3.1) and $C_{\mathrm{TEWS}}(T)$ is the vital-sign colour band. Thresholds follow the standard SATS implementation (TEWS 7 or more is Red): a single point difference can change the colour band, so nurses record each $f_k$ carefully. The diagram below visualises these bands on the score axis.

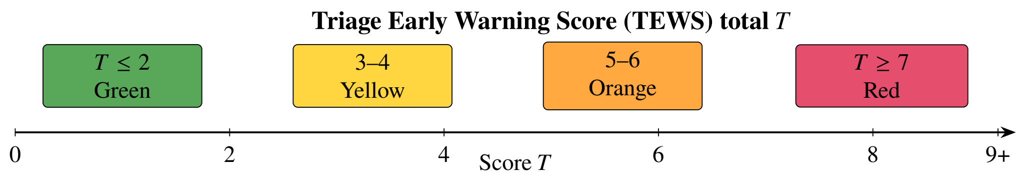


Figure 3.3: TEWS score ranges mapped to SATS colours (segments separated for readability).

### 3.3.4 Relationship to the text Softmax path

TEWS and BioBERT Softmax share a colour vocabulary (Red, Orange, Yellow, Green) but differ in inputs and mathematics:

- TEWS is a **weighted sum of seven vital features** $T = \sum_k w_k f_k(v_k)$ with $w_k = 1$ in the standard adult score, then a piecewise colour map $C_{\mathrm{TEWS}}(T)$.
- Softmax is a **Categorical distribution over five acuity levels** from complaint text, then the deterministic map $f_{\mathrm{SATS}}$.

They are complementary, not competitors. At arrival, text may be the only signal; after nurse assessment, TEWS and clinical discriminators remain authoritative. This project therefore documents both, implements Softmax fully, and never treats TEWS absence as a chatbot limitation: nurses already compute TEWS when vitals exist.

| **Aspect** | **TEWS** | **BioBERT Softmax** |
|---|---|---|
| Input | Seven vitals / flags $v_k$ (incl. SBP) | English chief complaint $X$ |
| Core map | Integer sum $T$ | Probabilities $\hat{\mathbf{y}}$ |
| Colour rule | $C_{\text{TEWS}}(T)$ bands | $f_{\text{SATS}}(\hat{c})$ |
| When available | After measurement | At text submission |
| Authority | Nurse desk standard | Intake assist (overridable) |

Table 3.4: Side-by-side comparison of TEWS and the text Softmax path.

| $k$ | Parameter | Measured $v_k$ | $f_k(v_k)$ | Running sum |
|---|---|---|---|---|
| 1 | HR | 125 bpm | 2 | 2 |
| 2 | RR | 26 breaths/min | 2 | 4 |
| 3 | SBP | 120 mmHg | 0 | 4 |
| 4 | Mobility | walking | 0 | 4 |
| 5 | Temperature | 36.8 °C | 0 | 4 |
| 6 | AVPU | alert | 0 | 4 |
| 7 | Trauma | absent | 0 | 4 |
| | | | **Total $T$:** | **4** |

Table 3.5: Term-by-term TEWS calculation for HR = 125 bpm, RR = 26 breaths/min, SBP = 120 mmHg, all other parameters normal.

**Example 3.1.** *Suppose vitals are HR* = 125 *bpm, RR* = 26 *breaths/min, and SBP* = 120 *mmHg; mobility, temperature, AVPU, and trauma are normal or absent. Table 3.5 lists each $f_k$ contribution; summing gives*

$$\begin{aligned}
f_1(125) &= 2 && \textit{(HR band } 111\text{–}129),\\
f_2(26) &= 2 && \textit{(RR band } 21\text{–}29),\\
f_3(120) &= 0 && \textit{(SBP band } 101\text{–}170),\\
f_k(\cdot) &= 0 && k = 4, \ldots, 7,\\
T &= 2 + 2 = 4, \\
C_{\text{TEWS}}(4) &= \textit{Yellow} \quad \textit{(Equation (3.5))}.
\end{aligned}$$

*If only these vitals were measured, the score is partial but still auditable. At intake, before vitals exist, the BioBERT path in Section 3.13 supplies the text-based colour.*

## 3.4 Neural networks, Transformers, and BERT

The text path of this project is implemented as a **neural network**: a composition of learnable maps that send numerical vectors to numerical vectors. Each map (a layer) has weights adjusted during training so that the overall composition predicts the correct acuity level from complaint text. Classical rule lists cannot capture the variety of English phrasing at intake; stacked nonlinear layers with attention can.

In the triage setting the input is already discrete after tokenisation (Section 3.7): integer IDs indexing an embedding table. The network never "reads English letters" directly; it reads vectors in $\mathbb{R}^{768}$ and repeatedly transforms them. Training adjusts those transformations so that the final Softmax puts high probability on the nurse-assigned acuity level.

A **Transformer** (Vaswani et al., 2017) is a neural architecture whose central mechanism is *self-attention*: every token in a sequence can attend to every other token in one parallel step. Unlike recurrent networks that read left-to-right only, a Transformer encoder builds context from the full window of tokens at once. That property matters for triage language, where negation and severity modifiers ("no pain", "severe headache") must reshape the meaning of neighbouring words. Scaled dot-product attention (Section 3.12) is the concrete formula used inside each BioBERT layer.

**BERT** (Bidirectional Encoder Representations from Transformers) (Devlin et al., 2019) is a deep Transformer *encoder* pre-trained with masked language modelling so that each token representation depends on left and right context. Masked language modelling hides random tokens and asks the model to recover them from surrounding biomedical or general text; the result is a reusable encoder rather than a chat generator. **BioBERT** (Lee et al., 2020) continues that pre-training on PubMed and PubMed Central text, then this project fine-tunes the encoder for five-class acuity prediction (Chapter 4).

Why BioBERT rather than a general BERT checkpoint alone? Domain continued pre-training places clinical vocabulary ("dyspnoea", "malaria", "epigastric") in a representation space already shaped by biomedical corpora, so Stage 3 fine-tuning on chief complaints starts closer to triage language than a purely Wikipedia-trained encoder would.

The remainder of this chapter specifies the pipeline in order: English input $X$, clinical gate, WordPiece tokenisation $\tau$, three embedding components, twelve encoder layers, Softmax classification over mutually exclusive SATS levels, colour map $f_{\text{SATS}}$, and pathway card $P(C)$.

## 3.5 Introducing BERT

TEWS ignores language. A patient may describe symptoms in free text before HR or RR are recorded. Subjective complaints often appear first; bidirectional encoding is required so that "not feverish" and "very feverish" are not treated alike.

Vaswani et al. (2017) introduced scaled dot-product attention (Section 3.12). Devlin et al. (2019) stacked attention into deep encoders with MLM pre-training. The input representation preview is

$$E(t_i) = E_{\text{word}}(t_i) + E_{\text{pos}}(i) + E_{\text{seg}}(s), \tag{3.6}$$

where $t_i$ is the token at position $i$, $E_{\text{word}}(t_i)$ identifies that token, $E_{\text{pos}}(i)$ encodes position $i$, and $E_{\text{seg}}(s)$ marks sentence segment $s$ (always $s = 0$ for a single complaint). The three vectors have the same length and are added **element-wise**, not concatenated: meaning, order, and segment combine into one vector per token. Full detail appears in Section 3.9.

| **Term** | **Meaning** | **Example in triage** |
|---|---|---|
| $E_{\text{word}}$ | Token identity: which word or subword? | "headache" vs "feverish" |
| $E_{\text{pos}}(i)$ | Position in the sequence: order matters | "head ache" $\neq$ "ache head" |
| $E_{\text{seg}}(s)$ | Sentence segment $s$ (A or B in paired inputs) | Always $s = 0$ for one complaint |

Table 3.6: Breakdown of the three terms in Equation (3.6).

**Remark 3.3.** *BERT processes the full sentence simultaneously. The representation of "feel feverish" depends on "headache" and neighbouring tokens in both directions, which is essential for triage wording.*

## 3.6 Introducing BioBERT

BioBERT reuses BERT's architecture (12 layers, 768 hidden dimensions, 12 attention heads) but continues MLM on biomedical corpora. Fine-tuning on nurse-labeled chief complaints adjusts the classification head so $h_{[\text{CLS}]}$ maps to five acuity levels.
Clinical tokens such as "headache", "feverish", "aches", and "weak" occupy regions of embedding space nearer related medical terms than in general BERT. After fine-tuning, their joint attention pattern supports acuity estimation for any presenting complaint, not only the illustrative example used in this chapter.

**Remark 3.4.** *Hospital intake is frequently text-first. BioBERT provides the mathematical bridge from unstructured complaint text to the same colour vocabulary nurses use with TEWS and SATS.*

## 3.7 Tokenisation

### 3.7.1 Input $X$

The NLP path begins with a single English *chief complaint* string $X$: the primary reason the patient presents, in free text. It is not a diagnosis, not a TEWS vital-sign vector, and not a full medical history. Non-clinical messages may be rejected by the relevance gate (Section 3.17) before tokenisation runs. Only clinical $X$ proceeds into the encoder.

### 3.7.2 The tokenisation map $\tau$

Neural networks require fixed-size numerical input. Tokenisation is the first transformation after the gate: a map $\tau$ sends raw complaint text $X$ into a bounded sequence of subword tokens with integer IDs.

$$X \xrightarrow{\tau} (t_1, \dots, t_M), \quad M \leq 128. \tag{3.7}$$

Here $X$ is the English chief-complaint string, $\tau$ is the **WordPiece tokenisation map** used by BERT and BioBERT, $t_1, \dots, t_M$ are the resulting subword tokens (with special markers), and $M$ is the sequence length capped at 128. The map $\tau$ is a *deterministic discrete encoding* that segments the string using a fixed vocabulary and emits token IDs. It is **not** a neural network layer; it does not learn during fine-tuning. Learning begins at the embedding lookup that follows.

### 3.7.3 Why the maximum length is $M = 128$

Native BioBERT / BERT-base supports sequences up to length 512. This project **deliberately uses** maximum sequence length $M = 128$ tokens for Stage 3 fine-tuning and chatbot inference. The shorter cap has three roles:

- **Batching.** Padding every complaint to length 128 yields rectangular tensors so many requests can share the same matrix multiply at intake.
- **Bound on context.** At most 128 subword positions (including `[CLS]`, `[SEP]`, and pads) enter one forward pass; longer free text must be truncated before submission.
- **Latency.** Shorter sequences reduce attention cost relative to the native 512 limit, which matters for sub-second intake response.

**Remark 3.5.** *The fixed length $M = 128$ is a* project deployment choice*, not a claim that BioBERT cannot handle longer inputs: the architecture default remains 512, while this chatbot truncates and pads to 128 for batching and latency.*

**WordPiece** keeps common clinical words ("fever", "pain") whole and splits rare or complex forms (e.g. "headache" → `head+##ache`). Vocabulary $\mathcal{V}$ has $|\mathcal{V}| \approx 28{,}996$ entries; each token ID indexes a row in the embedding table. When a word splits into two subwords, it occupies **two positions** in the sequence (see rows 4 and 7 in Table 3.8), so $M$ counts subword tokens, not raw words.

How the WordPiece algorithm works in detail. BioBERT ships with a fixed vocabulary of character and subword pieces learned offline; Stage 3 fine-tuning does not rebuild that vocabulary. At inference, each whitespace-separated word in $X$ is segmented by a left-to-right greedy longest-match search: starting at the first character, the algorithm takes the longest vocabulary entry that matches a prefix of the remaining string, emits that piece, then continues on the rest of the word. Continuation pieces after the first are marked with the prefix `##` so the model knows they belong to the same original word (for example `head` then `##ache`). If no vocabulary entry matches, the algorithm falls back to smaller pieces, ultimately to individual characters when necessary. Each emitted piece is then replaced by its integer ID in $\mathcal{V}$. Common clinical words such as "fever" or "pain" usually remain one token; rarer compounds such as "headache" or "feverish" split, which keeps the vocabulary manageable while still representing informal chief-complaint spelling. Those IDs are exactly the indices looked up in the embedding matrix in the next stage.

Three special tokens structure every input sequence:

- `[CLS]` (ID 101) is inserted at position $i = 0$ as the classification summary token; its final hidden state feeds the acuity head (Section 3.8).
- `[SEP]` (ID 102) is the **separator** token marking the end of the complaint sentence. It tells the model where clinical text stops, so subsequent <PAD> fillers are not interpreted as symptoms.
- <PAD> (ID 0) fills unused positions up to length $M = 128$; the attention mask sets these positions to zero so they do not affect encoding.

| **Token** | **ID** | **Purpose** |
|---|---|---|
| [CLS] | 101 | Classification summary; always first |
| [SEP] | 102 | End-of-sentence boundary before padding |
| <PAD> | 0 | Batch padding to fixed length 128 |

Table 3.7: Special tokens in the BioBERT tokenizer.

| $i$ | Raw | Token | ID |
|---|---|---|---|
| 0 | n/a | [CLS] | 101 |
| 1 | I | i | 1045 |
| 2 | have | have | 2031 |
| 3 | a | a | 1037 |
| 4 | headache | head+##ache | 7994, 8772 |
| 5 | and | and | 1998 |
| 6 | feel | feel | 2514 |
| 7 | feverish | fever+##ish | 9643, 6804 |
| 8 | . | . | 1012 |
| 9 | n/a | [SEP] | 102 |
| 10–127 | n/a | <PAD> | 0 |

Table 3.8: Token IDs for the illustrative complaint (padding to length 128 omitted).

## 3.8 The [CLS] and [SEP] tokens

Sequence classification requires one fixed-length vector summarising the entire complaint. Rather than averaging all token rows (which would weight filler words equally with symptoms), BERT introduces an artificial summary token [CLS] at position $i = 0$. The [SEP] token at the end of the real text marks the sentence boundary: encoding stops treating tokens after [SEP] as part of the clinical complaint, which matters once <PAD> fillers occupy the remaining positions. Only the [CLS] row is passed to the classification head; [SEP] itself is not used for acuity prediction.
After $L = 12$ encoder layers:

$$h_{[\mathrm{CLS}]} = H^{(L)}_{0,:} \in \mathbb{R}^{768}, \tag{3.8}$$

where $H^{(L)}$ is the final hidden-state matrix after all twelve layers and $H^{(L)}_{0,:}$ denotes its first row (the [CLS] position).

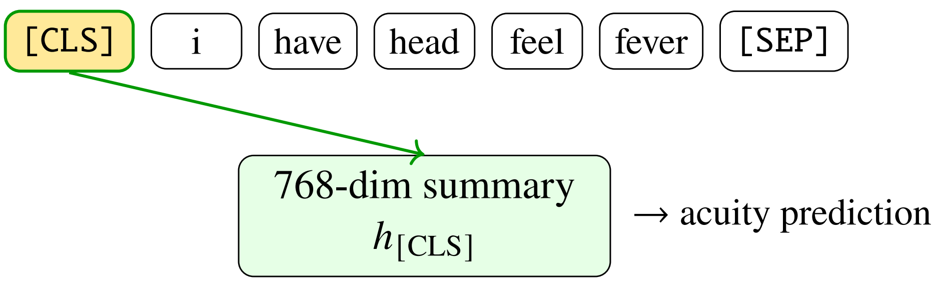


Figure 3.4: [CLS] aggregates context into one summary vector for classification.

Self-attention updates the [CLS] row at every layer so that it encodes information from all symptom tokens; the classification head therefore reads a single vector that represents the full clinical wording of $X$.

## 3.9 Embedding lookup

A token ID is an index, not a meaning. The embedding layer maps each ID to a dense vector in $\mathbb{R}^{768}$ via the learned matrix $E_{\text{word}} \in \mathbb{R}^{V \times 768}$:

$$\mathbf{e}_i = E_{\text{word}}[\text{id}(t_i)] \in \mathbb{R}^{768}, \tag{3.9}$$

where $t_i$ is the token at position $i$ and $\text{id}(t_i)$ is its integer vocabulary index (the WordPiece ID in Tables 3.7–3.8).

| Symbol | Definition |
|---|---|
| $V = \lvert\mathcal{V}\rvert$ | Vocabulary size (≈ 28,996) |
| $\text{id}(t_i)$ | Vocabulary index of token $t_i$ |
| $D = 768$ | Hidden dimension (BioBERT base configuration) |
| $\mathbf{e}_i$ | Word embedding vector for token at position $i$ |

Table 3.9: Embedding lookup notation.

The lookup $\mathbf{e}_i$ is one term in the full input representation $E(t_i) = E_{\text{word}} + E_{\text{pos}} + E_{\text{seg}}$ from Equation (3.6). After PubMed pre-training, BioBERT places clinically related terms nearer in this space than general-purpose embeddings would.

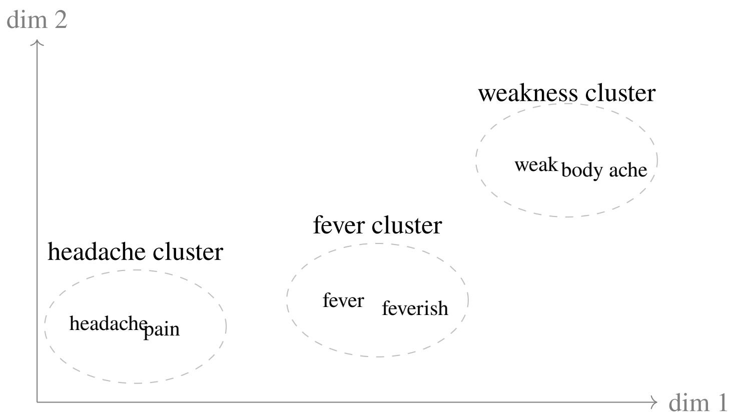


Figure 3.5: Schematic of BioBERT embedding space: familiar symptom words group together after biomedical pre-training (two examples per cluster).

When a patient writes "feverish" instead of "fever", or "head ache" as two tokens, the model can still map the wording into regions of this space associated with similar presentations. That generalisation across varied patient language is central to text-first triage.

Contextual mapping sums three embeddings (Equation (3.6)). The positional embedding depends only on index $i$, and the segment embedding depends on segment id $s$ (always

$s = 0$ for a single complaint):

$$E_{\text{pos}}(i) \in \mathbb{R}^{768}, \qquad E_{\text{seg}}(s) \in \mathbb{R}^{768}, \quad s = 0. \tag{3.10}$$

The final input vector for token $t_i$ at position $i$ is the element-wise sum across all $D = 768$ dimensions:

$$E(t_i)_d = E_{\text{word}}(t_i)_d + E_{\text{pos}}(i)_d + E_{\text{seg}}(s)_d, \quad d = 1, \dots, 768. \tag{3.11}$$

In words: for each dimension $d$, the word number, the position number, and the segment number are added together. The result is still a single 768-length vector for that token, not three vectors glued end to end.

### 3.9.1 Why three embedding components

The model does not use a single lookup. Each token must pass through **three** additive components before the encoder stack, for distinct reasons:

1. **Word embedding $E_{\text{word}}$.** Encodes *what* the subword is (clinical meaning of “feverish” versus “football”).

2. **Position embedding $E_{\text{pos}}(i)$.** Encodes *where* the token sits in the complaint. Transformers have no built-in left-to-right recurrence; without position, “severe pain then mild” could not be distinguished from the reverse order.

3. **Segment embedding $E_{\text{seg}}(s)$.** Marks sentence A versus B in paired BERT tasks. For a single chief complaint we always use $s = 0$; the term is kept so the BioBERT checkpoint remains architecturally compatible even though a second segment is unused at intake.

Omitting word embeddings would remove meaning; omitting position would scramble order-sensitive clinical phrasing; omitting segment would break compatibility with the pre-trained BERT/BioBERT weight layout. The three components are therefore not decorative: they are the minimal input interface the Transformer expects.

### 3.9.2 Why the hidden dimension is $D = 768$

BioBERT-base uses hidden width $D = 768$. That width is fixed by the published architecture (Lee et al., 2020; Devlin et al., 2019) and matches

$$12 \text{ attention heads} \times 64 \text{ dimensions per head} = 768. \tag{3.12}$$

Every token vector, every encoder layer, and the classification summary $h_{[\mathrm{CLS}]}$ live in $\mathbb{R}^{768}$. The width is large enough to encode rich biomedical semantics yet small enough to stack twelve layers and train on large corpora. Changing $D$ would require a different model family; this project therefore keeps $D = 768$ throughout.

| **Symbol** | **Definition** |
|---|---|
| $E_{\mathrm{word}}(t_i)$ | Row of $E_{\mathrm{word}}$ looked up by token ID |
| $E_{\mathrm{pos}}(i)$ | Learned vector for position $i$ in the sequence |
| $E_{\mathrm{seg}}(s)$ | Learned vector for segment $s$ (here $s = 0$) |
| $E(t_i)_d$ | Component $d$ of the summed embedding |

Table 3.10: Symbols in the full input embedding (Equation (3.11)).

**Example 3.2.** *For token "head" at position $i = 4$, suppose the first three dimensions (illustrative only) are*

$$\begin{aligned}
E_{\mathrm{word}} &= [0.31,\ -0.18,\ 0.52],\\
E_{\mathrm{pos}}(4) &= [0.05,\ 0.02,\ -0.01],\\
E_{\mathrm{seg}}(0) &= [0.00,\ 0.00,\ 0.00],\\
E(t_4) &= E_{\mathrm{word}} + E_{\mathrm{pos}}(4) + E_{\mathrm{seg}}(0)\\
&= [0.36,\ -0.16,\ 0.51] \quad \textit{(first three dimensions)}.
\end{aligned}$$

*The same element-wise addition repeats for all remaining dimensions $d = 4, \ldots, 768$. The result is row 4 of $H^{(0)}$.*

**Remark 3.6.** *The embedding table is like a dictionary: each token ID points to a list of 768 numbers encoding meaning. Adding position tells the model where the word appears; adding segment (zero here) reserves the mechanism for two-sentence inputs in other BERT tasks.*

Figure 3.6 shows the flow for one token position. The 768 dimensions are a fixed width chosen in the original BERT architecture: wide enough to encode rich semantics, yet small enough to train twelve layers on large corpora.

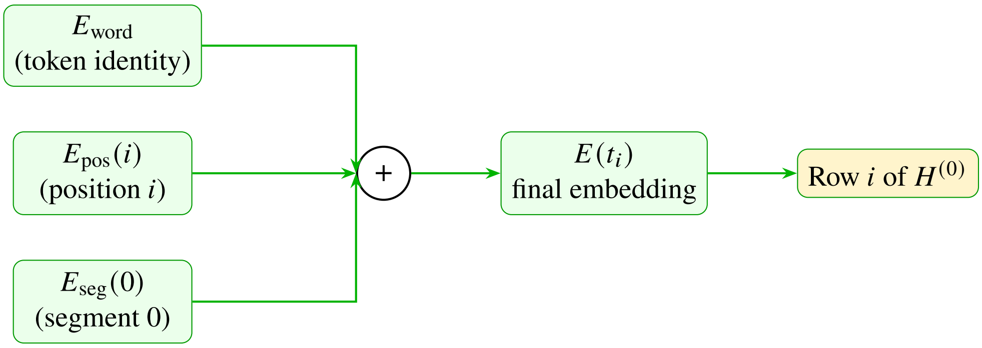


Figure 3.6: Three embeddings summed element-wise for one token position.

## 3.10 Input matrix $H^{(0)}$

Stacking the embedding vector for each of the $M$ tokens forms the initial hidden-state matrix

$$H^{(0)} \in \mathbb{R}^{M\times 768}, \tag{3.13}$$

where $M \le 128$ is the padded sequence length and 768 is the BioBERT hidden dimension. **Rows versus columns.** By convention in this project:

- **Rows** ($M \le 128$) index **token positions** in the sequence (row 0 is `[CLS]`, then the WordPiece tokens, then `[SEP]` and pads).
- **Columns** (768) index **hidden feature dimensions** of the BioBERT representation.

Row $i$ is built by applying Equation (3.11) at position $i$: look up the word vector, add the positional vector $E_{\text{pos}}(i)$, add the segment vector $E_{\text{seg}}(0)$, and stack the result. Viewed this way, a chief complaint is a table of numbers that the encoder layers progressively transform. The same row/column convention holds for $H^{(\ell)}$ after each encoder layer: shape stays $M \times 768$; only the entries change as self-attention mixes information *across rows*.

Later, the classification weight matrix $W \in \mathbb{R}^{5\times 768}$ uses the complementary layout: **rows** are the five acuity classes and **columns** are the 768 features that multiply $h_{[\text{CLS}]}$ (Section 3.13).

| $i$ | **Token** | $D_1$ | $D_2$ | $D_3$ | $D_4$ | $\cdots$ |
|---|---|---|---|---|---|---|
| 0 | `[CLS]` | 0.02 | 0.11 | −0.05 | 0.33 | $\cdots$ |
| 1 | i | −0.08 | 0.04 | 0.19 | 0.01 | $\cdots$ |
| 2 | have | 0.15 | −0.22 | 0.07 | 0.44 | $\cdots$ |
| 4 | head | 0.31 | −0.18 | 0.52 | 0.09 | $\cdots$ |
| 7 | fever | 0.44 | 0.27 | −0.11 | 0.61 | $\cdots$ |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ |

Table 3.11: Example submatrix of $H^{(0)}$ (first four of 768 dimensions; illustrative values).

After encoding, row 0 (`[CLS]`) holds the summary passed to the softmax classifier; the remaining rows are not used directly for the final colour decision.

## 3.11 Encoder layers

BioBERT applies the same encoder block $L = 12$ times. Each block first lets tokens attend to one another (mixing symptom phrases), then applies a feed-forward network (FFN) to each row independently. **Residual connections** add the block input back to its output so that gradients can flow through depth during training. **Layer normalisation** (LayerNorm) rescales each row to zero mean and unit variance for numerical stability (Vaswani et al., 2017; Devlin et al., 2019). MultiHeadAttention is the twelve-head self-attention block of Section 3.12. GELU is the Gaussian Error Linear Unit used in BERT/BioBERT FFNs (smooth nonlinear activation; not ReLU). The update equations are

$$
\begin{aligned}
Z^{(\ell)} &= \mathrm{LayerNorm}\big(H^{(\ell-1)} + \mathrm{MultiHeadAttention}(H^{(\ell-1)})\big), && (3.14)\\
H^{(\ell)} &= \mathrm{LayerNorm}\big(Z^{(\ell)} + \mathrm{FFN}(Z^{(\ell)})\big), && (3.15)\\
\mathrm{FFN}(x) &= W_2\,\mathrm{GELU}(W_1 x + b_1) + b_2, && (3.16)
\end{aligned}
$$

where $H^{(\ell-1)}$ and $H^{(\ell)}$ are the hidden-state matrices before and after layer $\ell$, $Z^{(\ell)}$ is the intermediate state after attention, and FFN uses $W_1 \in \mathbb{R}^{768\times3072}$, $W_2 \in \mathbb{R}^{3072\times768}$, and biases $b_1 \in \mathbb{R}^{3072}$, $b_2 \in \mathbb{R}^{768}$ in the standard BERT-base configuration. The inner dimension 3072 is four times the hidden width 768, giving the FFN enough capacity to transform each token row nonlinearly. In plain terms, each layer updates every token using the whole complaint, while the residual connection keeps the previous representation so meaning is refined rather than replaced.

| **Component** | **What it does** |
|---|---|
| MultiHeadAttention | Twelve parallel self-attention heads; each learns different word relationships |
| FFN | Two linear layers with GELU: expands each row to 3072 dims, then projects back to 768 |
| LayerNorm | Rescales activations for stable training across depth |
| Residual (+) | Adds sub-layer input to output so earlier information is not lost |

Table 3.12: Components inside one BioBERT encoder layer.

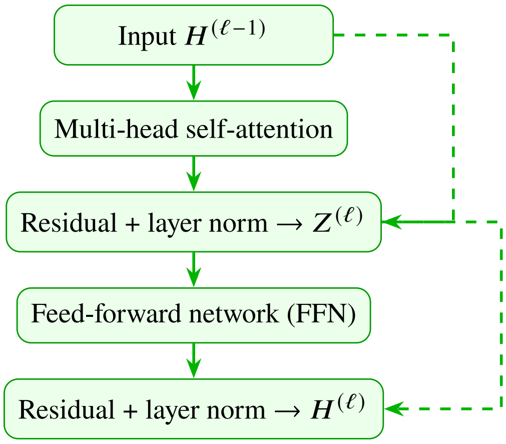


Figure 3.7: One BioBERT encoder layer (dashed arrows: residual connections). The block is stacked $L = 12$ times.

In the attention sub-step, each token row is updated using information from every other row in the complaint, which is how "headache" and "feverish" influence one another before acuity is estimated. The FFN sub-step then applies the same nonlinear transformation to each row separately, increasing representational capacity. Repeating this block twelve times transforms $H^{(0)}$ into $H^{(12)}$:

$$H^{(0)} \xrightarrow{\text{Layer 1}} H^{(1)} \xrightarrow{\text{Layer 2}} \cdots \xrightarrow{\text{Layer 12}} H^{(12)} = H^{(L)}.$$

Shallow layers (1–4) tend to encode individual words and short phrases; middle layers (5–8) combine symptoms such as "headache" with "feverish" and "weak"; deep layers (9–12) encode broader urgency patterns. All $M$ token rows are processed in parallel at each layer, not left-to-right as in older language models.

**Remark 3.7.** *Stacking layers is like reading a complaint once for words, once for symptom combinations, and once for overall urgency. The `[CLS]` row at layer 12 becomes $h_{[\mathrm{CLS}]}$ for classification.*

## 3.12 Self-attention

Self-attention computes a weighted combination of token representations for each position. The computation proceeds in four steps, matching Figure 3.8:

1. **Project** hidden rows into query, key, and value matrices.
2. **Score** pairwise compatibility between tokens.
3. **Normalise** scores into attention weights.
4. **Combine** value rows into context-aware outputs.

Let $H = H^{(\ell-1)} \in \mathbb{R}^{M\times 768}$ be the input to the current encoder layer. Learned projection matrices $W_Q, W_K, W_V \in \mathbb{R}^{768\times d_k}$ with key dimension $d_k = 64$ produce

$$\begin{aligned} Q = HW_Q, \quad K &= HW_K, \quad V = HW_V, \\ S &= QK^\top/\sqrt{d_k}, \\ A &= \mathrm{softmax}(S), \\ \mathrm{output} &= AV, \end{aligned}$$

where $Q, K, V \in \mathbb{R}^{M\times d_k}$, $S$ and $A$ are $M \times M$ score and weight matrices, and softmax is applied row-wise so each query distributes weights that sum to 1.
Following Vaswani et al. (2017), one attention head uses:

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \tag{3.17}$$

with key dimension $d_k = 64$ per head. For one query row, the softmax over keys assigns weight $\alpha_j$ to token $j$:

$$\alpha_j = \frac{\exp(e_j)}{\sum_{k=1}^{M}\exp(e_k)}, \quad \sum_{j=1}^{M}\alpha_j = 1, \tag{3.18}$$

where $e_j$ is the scaled score in row $j$ of $QK^\top/\sqrt{d_k}$ and $M$ is the sequence length (at most 128). BioBERT runs twelve such heads in parallel; their outputs are concatenated and projected back to 768 dimensions so each head can specialise (e.g. symptom co-occurrence vs negation).

**Remark 3.8.** *In triage language: Q is what the current token looks for in the rest of the complaint; K is the label each other token shows; V is the meaning that is mixed in when the match is strong. Dividing by* $\sqrt{d_k}$ *keeps the match scores from becoming too large before Softmax. The Softmax inside Equation* (3.17) *only mixes* ***token*** *meanings; it is not the five-class Softmax that later chooses the colour.*

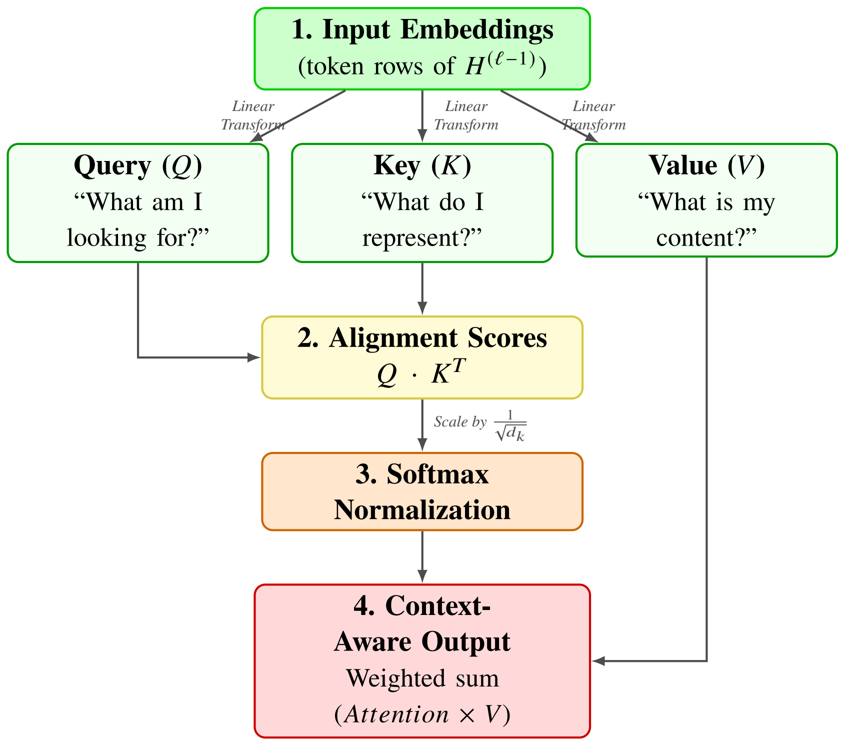


Figure 3.8: Self-attention data flow within one encoder layer.

| Symbol | Definition |
|---|---|
| $H = H^{(\ell-1)}$ | Hidden-state matrix entering the current encoder layer ($M \times 768$) |
| $W_Q, W_K, W_V$ | Learned projection matrices ($768 \times d_k$) that form queries, keys, and values |
| $Q, K, V$ | Query, key, and value matrices ($M \times d_k$) derived from $H$ |
| $QK^\top$ | Matrix of pairwise compatibility scores between tokens |
| $\sqrt{d_k}$ | Scaling constant ($d_k = 64$) that prevents dot products from growing too large |
| $e_j$ | Raw attention score for token $j$ before normalisation |
| $\alpha_j$ | Normalised attention weight for token $j$ (a probability) |
| $M$ | Sequence length (number of tokens, at most 128) |

Table 3.13: Self-attention notation.

| Query token | [CLS] | headache | feverish | filler |
|---|---|---|---|---|
| Attention to | 0.09 | 0.32 | 0.41 | 0.18 |

Table 3.14: Schematic attention weights when the query is [CLS] (illustrative, not measured from the model). Symptom tokens receive higher weight than filler words.

The product $QK^\top$ measures how strongly each token should attend to every other token. Dividing by $\sqrt{d_k}$ keeps those scores in a range where the softmax in Equation (3.18) does not saturate. In a complaint mentioning both "headache" and "feverish", attention typically assigns higher weight to symptom tokens than to function words such as “I” or “and”, so the updated representation emphasises clinically informative wording (Table 3.14).

# 3.13 Classification softmax

## 3.13.1 Mutually exclusive acuity categories

When a patient submits a chief complaint, their clinical urgency for routing is treated as falling into **exactly one** of the five SATS-aligned acuity levels $\{1, 2, 3, 4, 5\}$. A presentation cannot be Level 1 (immediate) and Level 5 (routine) at the same decision instant. Because only one level is selected for the pathway colour, the five categories are modelled as **mutually exclusive**. That modelling choice determines the output layer: the network must produce a distribution over five options, not five independent yes/no scores.

If the head used five independent sigmoid units, each class probability could be high at once (for example $\hat{y}_1 = 0.9$ and $\hat{y}_5 = 0.8$), which would not define a single routing colour. Softmax avoids that inconsistency by normalising across the five logits together.

## 3.13.2 Softmax as the probability translator

After BioBERT encoding, the classification head maps the summary vector $h_{[\mathrm{CLS}]} \in \mathbb{R}^{768}$ to five unconstrained real scores called **logits**:

$$\begin{aligned} \mathbf{z} &= W h_{[\mathrm{CLS}]} + \mathbf{b}, \\ \hat{\mathbf{y}} &= \mathrm{softmax}(\mathbf{z}), \\ \hat{y}_c &= \frac{\exp(z_c)}{\sum_{j=1}^{5} \exp(z_j)}, \end{aligned} \tag{3.19}$$

with $W \in \mathbb{R}^{5\times 768}$ (**rows** = acuity classes, **columns** = features), $\mathbf{b} \in \mathbb{R}^5$, logits $z_c$, and $\sum_{c=1}^{5} \hat{y}_c = 1$. Softmax enforces the two elementary rules of a finite probability distribution: every $\hat{y}_c$ is non-negative, and the five values sum to one. The predicted level is $\hat{c} = \arg\max_c \hat{y}_c$. Expressing output as probabilities allows downstream routing logic to consider confidence (for example a dominant Yellow at about 69% versus a near tie between Yellow and Green).

**Remark 3.9.** *The vector* $\mathbf{z}$ *holds five* ***raw scores (logits)****, not probabilities: a larger* $z_c$ *only means that urgency level looks stronger. Softmax turns those scores into five chances that add to 1. The largest chance is the acuity level used for the SATS colour.*

This **classification Softmax** (over five acuity classes) is distinct from the **attention Softmax** inside each encoder layer (over token positions). Both use the same functional form; they act on different axes of the model. Attention Softmax asks "which tokens matter for this query?"; classification Softmax asks "which acuity level is most probable for this complaint?"

**Remark 3.10.** *The linear map* $\mathbf{z} = Wh_{[\mathrm{CLS}]} + \mathbf{b}$ *is the triage analogue of a multi-class "score sheet": each row of* $W$ *emphasises features in* $h_{[\mathrm{CLS}]}$ *that favour one acuity level. Softmax then converts those raw scores into a comparable probability vector for hospital routing.*

### 3.13.3 Categorical distribution ($K = 5$)

By producing five probabilities that sum to one, Softmax parameterises a **Categorical distribution** over the acuity levels. A Categorical distribution is the discrete law for exactly $K$ mutually exclusive outcomes; here $K = 5$. The vector $\hat{\mathbf{y}}$ is the parameter of that law:

$$\hat{y}_c = P(\text{acuity} = c \mid X), \qquad c \in \{1, 2, 3, 4, 5\}, \tag{3.20}$$

where $X$ is the chief complaint and $\hat{y}_c$ is the Softmax probability of acuity level $c$. Informally, $\hat{\mathbf{y}}$ behaves like a weighted five-sided die whose face probabilities are determined by the complaint text. Sampling from that die would draw a random acuity; hospital routing instead takes the mode $\hat{c} = \arg\max_c \hat{y}_c$ (and may escalate to a nurse when the mode is not dominant).
Stage 3 training maximises the probability of the nurse-assigned level under this Categorical model. Equivalently, it minimises the negative log-likelihood (NLL)

$$\mathcal{L}_{\mathrm{NLL}} = -\log \hat{y}_y = -\log P(y \mid X), \tag{3.21}$$

where $y$ is the true acuity label, $X$ is the complaint, and $\hat{y}_y = P(y \mid X)$ is the Softmax probability of that label. This is exactly the cross-entropy loss used in Section 3.14 when labels are one-hot.

### 3.13.4 Bernoulli as the $K = 2$ building block

The **Bernoulli** distribution is the two-outcome special case ($K = 2$): a single trial with results $\{0, 1\}$ and success probability $p$. Softmax with two logits $(z_0, z_1)$ yields

$$p = \frac{\exp(z_1)}{\exp(z_0) + \exp(z_1)} = \sigma(z_1 - z_0), \tag{3.22}$$

where $z_0$ and $z_1$ are the two class logits, $p$ is the probability of class 1, and $\sigma(u) = 1/(1+e^{-u})$ is the logistic sigmoid. That identity shows why binary logistic regression is Softmax with $K = 2$. The five-class triage head is the same mathematical idea generalised from a weighted coin to five mutually exclusive faces.

**Example 3.3** (Bernoulli building block)**.** *Suppose a toy two-class head distinguishes "urgent" versus "non-urgent" with logits* $z_{\text{U}} = 1.0$ *and* $z_{\text{N}} = -0.5$*. Then*

$$\begin{aligned} \exp(z_{\text{U}}) &= e^{1.0} \approx 2.72, \\ \exp(z_{\text{N}}) &= e^{-0.5} \approx 0.61, \\ P(\text{urgent}) &= \frac{2.72}{2.72 + 0.61} \approx 0.82. \end{aligned}$$

*Raising* $z_{\text{U}}$ *relative to* $z_{\text{N}}$ *increases urgent probability exactly as raising one acuity logit relative to the others increases that class probability in the five-class Softmax.*

| **Symbol** | **Definition** |
|---|---|
| $h_{[\text{CLS}]}$ | 768-dimensional summary vector from final encoder layer |
| $W, \mathbf{b}$ | Learned classification weight matrix ($5 \times 768$) and bias |
| $z_c$ | Logit (pre-softmax score) for acuity level $c$ |
| $\hat{y}_c$ | Categorical probability of level $c \in \{1, \ldots, 5\}$ |
| $\hat{c}$ | Predicted level: index of largest $\hat{y}_c$ |

Table 3.15: Classification head notation.

**Example 3.4** (Five-class Softmax for the running complaint)**.** *Suppose illustrative logits for levels 1–5 are*

$$\begin{aligned} \mathbf{z} &= [-1.2,\ -0.5,\ 1.8,\ 0.3,\ -0.9], \\ \exp(\mathbf{z}) &\propto [0.30,\ 0.61,\ 6.05,\ 1.35,\ 0.41], \\ \sum_{c=1}^{5} \exp(z_c) &= 8.72, \\ \hat{\mathbf{y}} &\approx [0.03,\ 0.07,\ 0.69,\ 0.15,\ 0.05], \\ \hat{c} &= 3, \\ f_{\text{SATS}}(3) &= \textit{Yellow}. \end{aligned}$$

*Level 3 has the largest Categorical probability (*$\hat{y}_3 \approx 0.69$*, rounded to 72% in some deployment logs; Section 3.15). The mass on neighbouring Orange (*$0.07$*) and Green (*$0.15 + 0.05$*) shows residual uncertainty: Softmax does not claim certainty, only a ranked distribution suitable for pathway assist with nurse oversight when confidence is low.*

**Remark 3.11.** *Softmax probabilities are calibrated only to the extent that the training distribution matches live intake. High $\hat{y}_c$ on Triagegeist validation text is not a clinical guarantee at KNUST University Hospital; low-confidence or contradictory presentations remain mandatory nurse review.*

## 3.14 Model training pipeline

The deployed triage classifier is not trained from random initialisation. It is the product of three sequential stages: biomedical pre-training (Stage 1), encoder transfer from a prior triage checkpoint (Stage 2), and project fine-tuning on nurse-labeled chief complaints (Stage 3). Stages 1 and 2 supply medical language and partial triage adaptation; Stage 3 is the comprehensive retraining that maps free-text complaints to five acuity levels for this chatbot.

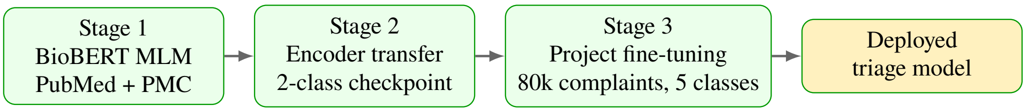


Figure 3.9: Three-stage training stack for the BioBERT triage classifier.

### 3.14.1 Stage 1: BioBERT pre-training (masked language modelling)

Before any triage labels are used, BioBERT is trained on biomedical text with masked language modelling (MLM). Random tokens are hidden and the model must predict them from surrounding context in both directions:

$$\mathcal{L}_{\mathrm{LM}} = -\sum_{i \in \mathcal{M}} \log P(t_i \mid H^{(L)}; \theta), \tag{3.23}$$

where $\mathcal{M}$ is the set of masked positions, $t_i$ is the true token at position $i$, $H^{(L)}$ is the final hidden-state matrix from all twelve layers, $\theta$ denotes all model parameters, and $P(t_i \mid \cdot)$ is the softmax probability assigned to the correct word. PubMed abstracts and PubMed Central full text ($\approx$ 18 billion words) teach medical vocabulary and syntax. Stage 1 does **not** teach triage urgency, SATS colours, or informal chief-complaint phrasing.

| Symbol | Definition |
|---|---|
| $\mathcal{M}$ | Set of masked token positions in the sentence |
| $t_i$ | True (correct) word at masked position $i$ |
| $\theta$ | All model weights and biases |
| $P(t_i \mid H^{(L)}; \theta)$ | Model probability for the correct token |
| $\mathcal{L}_{\text{LM}}$ | Total MLM penalty; lower means better predictions |

Table 3.16: Masked language modelling notation (Stage 1).

**Example 3.5.** *If the masked token should be "headache", the MLM contribution is*

*Good prediction ($P = 0.92$):*

$$-\log(0.92) \approx 0.08 \qquad \textit{(small penalty)},$$

*Poor prediction ($P = 0.01$):*

$$-\log(0.01) \approx 4.6 \qquad \textit{(large penalty)}.$$

### 3.14.2 Stage 2: Encoder transfer from a triage checkpoint

A publicly available BioBERT triage checkpoint partially adapted the encoder to urgency-related text using a **two-class** classification head ($W \in \mathbb{R}^{2\times768}$). For this project, that 2-class head is **discarded** on load because our labels require five acuity levels. The loader uses `ignore_mismatched_sizes=True`: only encoder weights carry forward; the output layer is randomly reinitialised for five classes. Stage 2 therefore contributes triage-oriented encoder features without constraining the final colour mapping.

### 3.14.3 Stage 3: Fine-tuning on nurse-labeled chief complaints

Stage 3 is the comprehensive retraining performed for this project. Nurse-assigned acuity labels from the Triagegeist benchmark (Section 3.1.2) are inner-joined with chief-complaint text on patient identifier, yielding $N = 80{,}000$ English text-label pairs $(X_i, y_i)$. Only the free-text English chief complaint (why the patient came) and the nurse-assigned urgency level (1–5) are used; vitals and demographics in the source records are excluded so the NLP path matches the single-shot text input at intake.

The data are split 90/10 into training and validation sets ($\approx$ 72,000 / $\approx$ 8,000 rows, `test_size=0.1`, random seed 42). A new classification head $W \in \mathbb{R}^{5\times768}$, $\mathbf{b} \in \mathbb{R}^5$ is trained from scratch while encoder weights are updated with a small learning rate.

| **Level** | **Count** | **Proportion** | **SATS colour** |
|---|---|---|---|
| 1 (most urgent) | 3,222 | 0.0403 | Red |
| 2 | 13,439 | 0.1680 | Orange |
| 3 | 28,921 | 0.3615 | Yellow |
| 4 | 23,020 | 0.2878 | Green |
| 5 (least urgent) | 11,398 | 0.1425 | Green |
| **Total** | **80,000** | **1.000** | |

Table 3.17: Class distribution in the 80,000 labeled chief-complaint training pairs.

| **Hyperparameter** | **Value** |
|---|---|
| Epochs | 3 |
| Batch size (train and eval) | 16 |
| Learning rate $\alpha$ | $2 \times 10^{-5}$ |
| Weight decay | 0.01 |
| Max sequence length | 128 tokens |
| Mixed precision (FP16) | Enabled |
| Total optimizer steps | 13,500 (3 epochs $\times$ 4,500 steps) |
| Best validation loss | 0.001812 (epoch 3) |

Table 3.18: Fine-tuning hyperparameters (Stage 3).

| **Component** | **Updated during Stage 3?** |
|---|---|
| Embedding layers | Yes (small learning-rate updates) |
| 12 transformer encoder layers | Yes |
| Classification head $W$, $\mathbf{b}$ | Yes (trained from scratch) |

Table 3.19: Which model components change during project fine-tuning.

The objective is **cross-entropy loss**: for each training example, the model is penalised when it assigns low probability to the nurse's true acuity label. For a single example with true class $y$,

$$\mathcal{L}_{\mathrm{CE}} = -\log \hat{y}_y, \tag{3.24}$$

where $y \in \{1, \ldots, 5\}$ is the nurse-labeled acuity level and $\hat{y}_y$ is the Softmax probability the model assigns to that correct level. Over the full training set the same symbol denotes the empirical risk

$$\mathcal{L}_{\mathrm{CE}} = -\sum_{i=1}^{N} \log P(y_i \mid X_i;\, \theta), \quad N = 80{,}000, \tag{3.25}$$

where $X_i$ is complaint $i$, $y_i$ is its label, $\theta$ denotes all trainable parameters, and $P(y_i \mid X_i; \theta) = \hat{y}_{y_i}$ is the predicted probability of the true class.

**Example 3.6.** *If the nurse labeled a complaint Yellow (level 3), the cross-entropy penalty is*

*Low confidence on the true class ($\hat{y}_3 = 0.05$):*

$$\mathcal{L}_{\mathrm{CE}} = -\log(0.05) \approx 3.0 \qquad \textit{(large penalty)},$$

*High confidence on the true class ($\hat{y}_3 = 0.95$):*

$$\mathcal{L}_{\mathrm{CE}} = -\log(0.95) \approx 0.05 \qquad \textit{(small penalty)}.$$

*Fine-tuning adjusts weights so correct classes receive higher probability on similar complaints.*

**Backpropagation** computes how each weight contributed to $\mathcal{L}_{\mathrm{CE}}$ by applying the chain rule backward from the loss through the softmax head and all twelve encoder layers. For classification weights $W \in \mathbb{R}^{5\times768}$, with logits $\mathbf{z} = Wh_{[\mathrm{CLS}]} + \mathbf{b}$ where $h_{[\mathrm{CLS}]} \in \mathbb{R}^{768}$ is the final summary vector and $\mathbf{b} \in \mathbb{R}^5$ is the bias,

$$\frac{\partial \mathcal{L}_{\mathrm{CE}}}{\partial W} = \frac{\partial \mathcal{L}_{\mathrm{CE}}}{\partial \hat{\mathbf{y}}} \cdot \frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{z}} \cdot \frac{\partial \mathbf{z}}{\partial W}. \tag{3.26}$$

In words: (i) $\partial \mathcal{L}_{\mathrm{CE}}/\partial \hat{\mathbf{y}}$ measures how wrong the predicted probabilities are; (ii) $\partial \hat{\mathbf{y}}/\partial \mathbf{z}$ propagates through softmax; (iii) $\partial \mathbf{z}/\partial W$ links logits to the weight matrix. The same logic applies layer by layer through BioBERT.

**Gradient descent** then adjusts weights in the direction that reduces the loss:

$$W_{\mathrm{new}} = W_{\mathrm{old}} - \alpha \frac{\partial \mathcal{L}_{\mathrm{CE}}}{\partial W}, \tag{3.27}$$

where $\alpha = 2\times10^{-5}$ is the learning rate, $W_{\mathrm{old}}$ is the current weight matrix, and $W_{\mathrm{new}}$ is the updated matrix. When the model confidently predicts the wrong colour, backpropagation increases the loss sharply and the subsequent weight update pushes similar complaints toward the nurse-labeled class on the next training pass.

**Remark 3.12.** *Pre-training (Stage 1) gives the model medical language; comprehensive fine-tuning (Stage 3) teaches triage decisions on short, informal chief-complaint phrasing. The learning rate is kept small so PubMed knowledge is preserved while the head and encoder adapt to urgency labels.*

### 3.14.4 Mitigating class imbalance

Table 3.17 shows that Level 3 dominates Triagegeist while Level 1 (Red) is scarce. The published *stratified-oversampling* checkpoint therefore rebalances the **training** split only (the validation split is left untouched) before a second Stage 3 run, with embedding-space SMOTE disabled (`USE_EMBEDDING_SMOTE=False`). The algorithmic steps are:

1. Restrict imbalance mitigation to the stratified Triagegeist training rows from Section 3.14.3.
2. Apply light **text augmentation** to minority urgent classes (acuity levels 1 and 2): synonym or paraphrase variants of the chief-complaint string, with protected medical keywords left unchanged. Configured yields are two augmented variants per Level 1 row and one per Level 2 row.
3. Apply **random minority oversampling** with replacement until target counts $n_1$ = 10,000 and $n_2$ = 15,000; Levels 3–5 keep their natural training counts.
4. Run end-to-end Stage 3 fine-tuning (WordPiece → BioBERT → Softmax) on the expanded training set with the same hyperparameters as Table 3.18.

Chapter 4 contrasts this checkpoint with the unrebalanced baseline and explains why classical embedding SMOTE was not chosen (Section 4.2.4).

## 3.15 Mapping acuity to colour: $f_{\text{SATS}}$

The Softmax head predicts Emergency Severity Index (ESI)–style acuity levels 1–5 learned from training and external evaluation labels (Triagegeist, MIMIC-IV-ED, NHAMCS). Those sources are **not** SATS-labelled: ESI is primarily resource-based, whereas SATS combines vital-sign TEWS with clinical discriminators. Hospital routing at KNUST University Hospital nonetheless uses SATS colours, so this project defines a fixed deterministic operational heuristic $f_{\text{SATS}}$ that bridges the shared five-tier urgency curve to Ghanaian pathway colours:

$$f_{\text{SATS}}(\hat{c}) = C_{\text{NLP}} = \begin{cases} \text{Red} & \hat{c} = 1 \\ \text{Orange} & \hat{c} = 2 \\ \text{Yellow} & \hat{c} = 3 \\ \text{Green} & \hat{c} \in \{4, 5\} \end{cases} \tag{3.28}$$

where $\hat{c} = \arg\max_c \hat{y}_c$ is the predicted acuity level, $f_{\text{SATS}}$ is the fixed level-to-colour map, and $C_{\text{NLP}}$ is the resulting SATS colour from the text path.
While ESI and SATS operate on fundamentally different clinical philosophies (resource allocation versus physiological discriminators), they share an isomorphic five-tier urgency curve. The heuristic $f_{\text{SATS}}$ leverages this shared architecture to bridge text-based acuity predictions to Ghanaian pathway colours, functioning strictly as an antecedent

routing assist rather than a definitive clinical cross-walk, and not as a claim that the training or external datasets are SATS-labelled.

| ESI-style level | Clinical name | $f_{\text{SATS}}(\ell)$ |
|---|---|---|
| 1 | Resuscitation | Red |
| 2 | Emergent | Orange |
| 3 | Urgent | Yellow |
| 4 | Less urgent | Green |
| 5 | Non-urgent | Green |

Table 3.20: Proposed ESI-style acuity → SATS colour heuristic for text-first routing.

Levels 4 and 5 both map to Green because both are non-urgent in the five-level schema. Blue is reserved for dignity protocol on arrival and is not produced by this five-class head.

## 3.16 Clinical pathways $P(C_{\text{NLP}})$

Once $C_{\text{NLP}}$ is known, the chatbot attaches a pathway card that specifies waiting time, destination, and escalation rules drawn from SATS-aligned hospital protocol at KNUST University Hospital:

$$P(C_{\text{NLP}}) = \big(T_{\max},\ \text{destination},\ \text{actions},\ \text{escalation}\big), \tag{3.29}$$

where $P(C_{\text{NLP}})$ is the pathway card for colour $C_{\text{NLP}}$, $T_{\max}$ is the maximum waiting time, destination is the primary clinical department, actions list support steps (nursing, pharmacy, diagnostics), and escalation defines what happens if the timer expires or the patient worsens. Colour alone is not enough at intake: the pathway card names where the patient should go and how soon they should be seen. The four tuple components correspond directly to the columns in Table 3.21.

**Lexical child-marker rule.** BioBERT has no age feature, so Pediatrics is never a default destination for adult ASTS colours. Before the pathway card is finalised, a deterministic (non-neural) check runs on the complaint $X$. Define the marker set

$$\mathcal{M}_{\text{child}} = \{\texttt{child}, \texttt{son}, \texttt{daughter}, \texttt{baby}, \texttt{infant}, \texttt{toddler}\}. \tag{3.30}$$

Normalise $X$ to lowercase. If any element of $\mathcal{M}_{\text{child}}$ appears as a substring of $X$, override the destination field of $P(C_{\text{NLP}})$ to Pediatrics; otherwise keep the adult destination for that colour. This rule is applied after $C_{\text{NLP}} = f_{\text{SATS}}(\hat{c})$ and does not change the colour itself.

| Colour | $T_{\max}$ | Destination | Escalation |
|---|---|---|---|
| Red | 0 min | Resuscitation / Surgery | Immediate senior alert |
| Orange | 10 min | Internal Medicine / Surgery / Obs. & Gynae | Head nurse if unclaimed |
| Yellow | 60 min | Outpatient urgent stream | Re-assess if timer expires |
| Green | < 4 h | Outpatient / Dental / Eye / Public & Preventive Health | Return only if worse |

Table 3.21: SATS pathway protocol at KNUST University Hospital (clinical destinations; adult ASTS default).

Table 3.22 is the authoritative full map: one row per NLP outcome (including gate rejection), with primary clinical departments, support actions, escalation rules, and an example complaint phrase.

| Scenario | Trigger | $T_{\max}$ | Primary clinical departments | Support actions | Escalation |
|---|---|---|---|---|---|
| Gate reject | $g(X) < 0.30$ | n/a | Hold at intake | Nursing: request complaint; HIT: no colour stored | Re-submit with clinical text |
| Red | $\hat{c} = 1$ | 0 min | Surgery, Internal Medicine | Nursing (vitals, TEWS), Pharmacy (emergency), Health Info (Red flag), Admin (bed bypass), HIT (Code Red) | Immediate senior clinician |
| Orange | $\hat{c} = 2$ | 10 min | Internal Medicine, Surgery, Obs. & Gynae | Nursing (monitor, timer), Diagnostics (stat labs), Pharmacy, Health Info, Admin, HIT | Head nurse if unclaimed |
| Yellow | $\hat{c} = 3$ | 60 min | Outpatient urgent, Diagnostics | Nursing (vitals, TEWS, timer), Pharmacy, Health Info, HIT, Accounts (defer billing) | Re-assess at 60 min |
| Green | $\hat{c} \in \{4, 5\}$ | <4 h | Outpatient, Dental, Eye, Public & Preventive Health | Nursing (brief triage), Diagnostics (routine), Health Info, HIT, Accounts | Return if worse |

Table 3.22: Complete NLP pathway scenarios at KNUST University Hospital (gate reject plus four SATS colours). Pediatrics is not a default Yellow destination; destination is overridden to Pediatrics only when the lexical child-marker rule matches $\mathcal{M}_{\text{child}}$ in $X$.

| Scenario | Example complaint phrase |
|---|---|
| Gate reject | "Hello, how are you?" (no clinical content) |
| Red | "Crushing chest pain, cannot breathe, sweating heavily" |
| Orange | "Severe abdominal pain in pregnancy, bleeding" |
| Yellow | *"I have a headache and feel feverish. My body aches and I feel weak."* |
| Green | "Routine dental pain for two weeks, no fever" |

Table 3.23: Example chief complaints triggering each pathway scenario.

Table 3.24 maps all hospital departments to their role in the pathway system. Clinical departments receive patients by colour; support departments execute actions attached to each pathway card.

| Department | Role | Colours / function |
|---|---|---|
| Internal Medicine | Clinical | Red, Orange (acute medical) |
| Surgery | Clinical | Red, Orange (trauma, acute surgical) |
| Obstetrics and Gynaecology | Clinical | Orange (obstetric urgency) |
| Pediatrics | Clinical | Child markers only (not default Yellow) |
| Outpatient | Clinical | Yellow (urgent), Green (routine) |
| Diagnostics Services | Clinical | Yellow, Green (parallel labs/imaging) |
| Dental | Clinical | Green |
| Eye | Clinical | Green |
| Public Health | Clinical | Green (community referrals) |
| Preventive Health | Clinical | Green (screening, wellness) |
| Nursing | Support | All colours (triage, vitals, reassessment) |
| Pharmacy | Support | All colours (medication on protocol) |
| Health Information | Support | All colours (record colour and timer) |
| Hospital Information Technology | Support | All colours (digital routing, API) |
| Administration | Support | All colours (bed and queue management) |
| Accounts | Support | Green, Yellow (billing after stabilisation) |

Table 3.24: Department routing map at KNUST University Hospital.

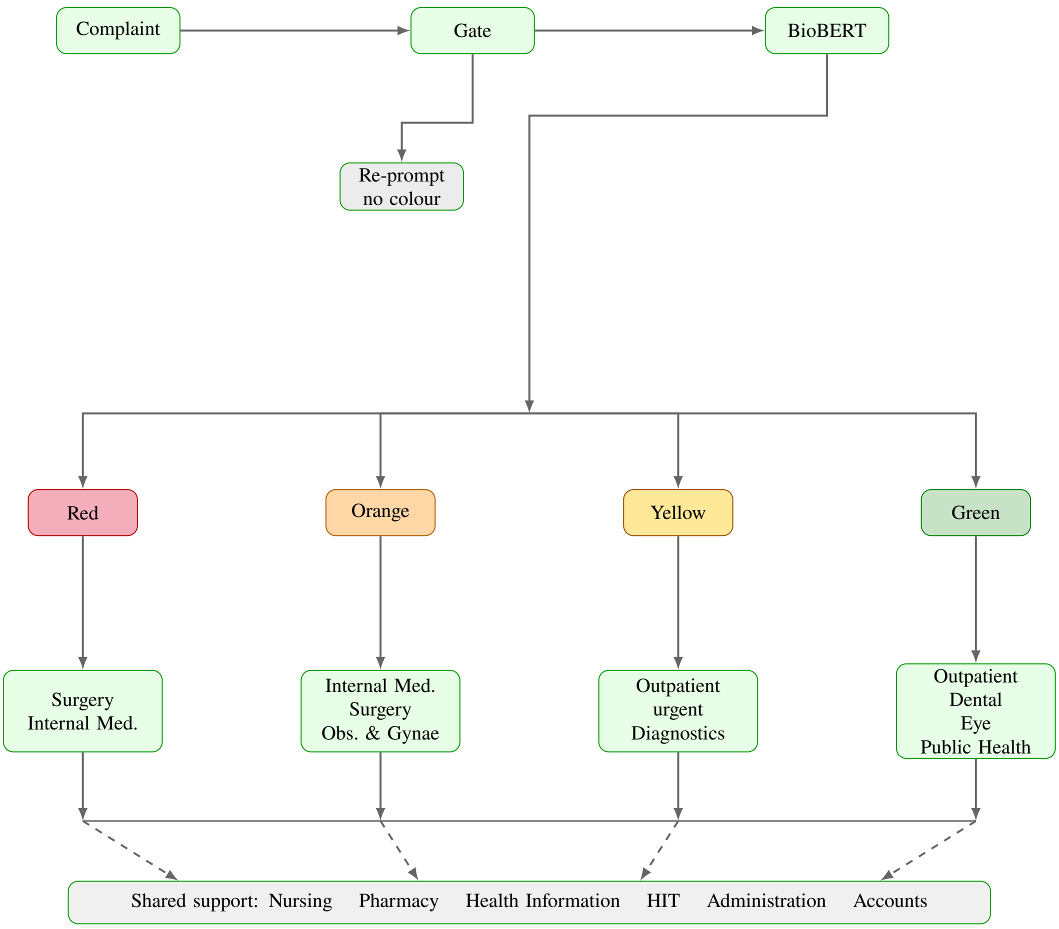


Figure 3.10: Complete NLP pathway map at KNUST University Hospital: gate rejection and all four SATS colours with clinical destinations and shared support services.

**Remark 3.13.** *Routing is protocol-driven; nurses may override the chatbot suggestion after clinical assessment. The chatbot supplies an initial colour and department direction before vitals are measured, reducing idle time at intake.*

**Gate rejection.** When $g(X) < 0.30$, the pipeline stops before BioBERT (Table 3.22). Nursing staff request a proper chief complaint; no colour or pathway card is issued and Health Information stores no triage record until valid clinical text is submitted.

**Red ($\hat{c} = 1$).** Immediate life-threatening presentations (e.g. crushing chest pain with dyspnoea) map to Red with $T_{\max} = 0$. Surgery and Internal Medicine activate resuscitation pathways; Nursing records immediate vitals and TEWS; Administration bypasses routine registration; Hospital Information Technology raises a Code Red alert.

**Orange ($\hat{c} = 2$).** Very urgent cases such as severe abdominal pain in pregnancy route to Internal Medicine, Surgery, or Obstetrics and Gynaecology within ten minutes. Nursing starts a countdown timer; Diagnostics may run stat laboratory work; escalation to the head nurse occurs if no clinician claims the patient within $T_{\max}$.

**Yellow ($\hat{c} = 3$).** The running illustrative complaint (*"I have a headache and feel feverish. My body aches and I feel weak."*) maps here at approximately 72% confidence. Outpatient urgent stream and parallel Diagnostics orders apply under the adult ASTS default; Pediatrics applies only when the lexical child-marker rule fires on $X$. Nursing sets a 60-minute reassessment timer and records the colour in Health Information.

**Green ($\hat{c} \in \{4, 5\}$).** Non-urgent presentations (e.g. routine dental pain without fever) stream to Outpatient, Dental, Eye, or Public and Preventive Health within four hours. This diverts low-acuity patients away from acute waiting areas; Accounts handles billing after brief Nursing triage.

## 3.17 Medical relevance gate

Running BioBERT on non-clinical text would waste compute and produce meaningless acuity scores. A lightweight rule-based gate scores clinical relevance before the deep model runs:

$$g(X) = \mathrm{clip}\left(\sum_j s_j(X) - p(X),\ 0,\ 1\right), \tag{3.31}$$

where

- $X$ is the English chief-complaint string;
- $\sum_j s_j(X)$ is an additive clinical evidence score assembled as follows (text-only path used in this project):
  - symptom lexicon: +0.15 per matched term, capped at 0.45;
  - triage or hospital-context phrases (e.g. OPD, nurse, chief complaint): +0.25 if present;
  - optional disease or drug entity bonuses when an external entity tagger is available;
- $p(X)$ is the non-clinical penalty: $p(X) = 0.5$ if a greeting-only, social-chatter, or other non-clinical pattern matches, and $p(X) = 0$ otherwise;
- $\mathrm{clip}(u, 0, 1) = \max(0, \min(1, u))$ keeps the gate score in the unit interval $[0, 1]$.

**Greeting and non-clinical patterns.** The gate does not treat "greeting" as a vague category. Standalone greeting-only strings matched by the prototype include `hi`, `hello`, `hey`, `good morning`, `good afternoon`, and `good evening` (optional `there`, optional trailing punctuation). Social chatter combines a greeting opener with `how are you` or `what's up`. Separate non-clinical topic patterns cover sports (e.g. playing football),

shopping, school work, weather, travel, work chat, cooking, and entertainment. Any such match sets $p(X) = 0.5$.
In order: add the clinical evidence, subtract the non-clinical penalty, then clip into $[0, 1]$. Classification proceeds only if

$$g(X) \geq \tau, \qquad \tau = 0.30, \tag{3.32}$$

where $\tau$ is the acceptance threshold (overridable by environment configuration in the prototype).

**Why $\tau = 0.30$.** With the lexicon weights above, one symptom term yields 0.15, two terms yield 0.30, and three terms yield 0.45 (before the 0.45 cap binds). Setting $\tau = 0.30$ therefore admits short but clear complaints such as "chest pain" (two lexicon hits) while still rejecting greeting-only and sports chat, which score $g(X) = 0$ after the penalty. The older candidate $\tau = 0.35$ sits above two lexicon hits, so it systematically rejects two-term clinical phrases; that raises false clinical rejection risk and was therefore not retained as the project default.

**Empirical gate validation.** The threshold choice is supported by a hand-labeled contrast set of $N = 52$ English strings (28 clinical, 24 non-clinical), including multi-symptom complaints, short clinical phrases, lexicon-poor clinical wording (e.g. "I feel something strange in my chest"), named greetings, and social or sports chat. Treating clinical text as the positive class, define

$$\text{Sens} = \frac{TP}{TP + FN}, \qquad \text{Spec} = \frac{TN}{TN + FP},$$

together with accuracy, precision, and the confusion counts $TP, FP, TN, FN$. Here $FN$ is a *false clinical rejection* (clinical text blocked before BioBERT) and $FP$ is a *false clinical acceptance* (non-clinical text allowed through). Table 3.25 summarises both thresholds on this set.

| $\tau$ | Acc | Prec | Sens | Spec | $TP$ | $FP$ | $TN$ | $FN$ |
|---|---|---|---|---|---|---|---|---|
| 0.30 (deployed) | 88.46% | 100% | 78.57% | 100% | 22 | 0 | 24 | 6 |
| 0.35 (candidate) | 61.54% | 100% | 28.57% | 100% | 8 | 0 | 24 | 20 |

Table 3.25: Clinical relevance gate metrics on the $N = 52$ hand-labeled contrast set. Positive class = clinical. $FN$ = false clinical rejection; $FP$ = false clinical acceptance.

At the deployed $\tau = 0.30$, specificity and precision are 100% on this set (no false clinical acceptances among the 24 non-clinical strings), while sensitivity is 78.57% (6 false clinical rejections). Raising $\tau$ to 0.35 leaves specificity unchanged but collapses sensitivity to 28.57% (20 false rejections), which is why 0.30 is preferred. Residual false rejections at 0.30 are mainly lexicon-poor clinical wording (e.g. "I feel something strange in my chest", "I am not feeling well") that never accumulate enough $s_j$. In those cases the gate fails closed for colour and asks for a richer chief complaint; nurse override and re-submission remain the safety net. This contrast set is illustrative, not a prospective audit of live KNUST University Hospital traffic.

| **Example input** $X$ | $g(X)$ | Gate | Outcome |
|---|---|---|---|
| "chest pain" | 0.30 | Pass | Classify |
| "chest pain and fever since this morning" | $\geq 0.30$ | Pass | Classify |
| "headache and feverish" | 0.45 | Pass | Classify |
| "I feel something strange in my chest" | 0.15 | Fail | Request clinical text |
| "Hello, how are you?" | 0 | Fail | Request clinical text |
| "good morning" | 0 | Fail | Request clinical text |
| "Going to play football" | 0 | Fail | Request clinical text |

Table 3.26: Illustrative clinical relevance gate outcomes at the deployed threshold $\tau = 0.30$.

**Example 3.7.** *For "headache and feverish", the lexicon matches three overlapping terms (*`headache`*,* `fever`*,* `ache`*), giving*

$$\sum_j s_j = \min(0.45,\ 3 \times 0.15) = 0.45,$$

$$g(X) = \mathrm{clip}(0.45 - 0,\ 0,\ 1) = 0.45 \geq 0.30,$$

*so classification proceeds. For "Hello, how are you?" with no symptom hits and greeting or social-chatter penalty $p(X) = 0.5$:*

$$\sum_j s_j = 0,$$

$$g(X) = \mathrm{clip}(0 - 0.5,\ 0,\ 1) = 0 < 0.30,$$

*and the system requests a chief complaint instead of running BioBERT.*

## 3.18 Worked numerical walkthrough

This section traces one complaint through every stage of the NLP pipeline in Figure 3.1 and Section 3.19 with concrete numbers, so the Softmax and pathway sections are not left abstract. Let

$$X = \text{“}\textit{I have a headache and feel feverish. My body aches and I feel weak.}\text{”}$$

as elsewhere in the project.

**Step 1: Gate.** Symptom lexicon hits for the running complaint include `headache`, `fever/ache`, and `weak`, giving $\sum_j s_j = \min(0.45,\ 4\times 0.15) = 0.45$ and $g(X) = 0.45 \geq 0.30$ with no greeting penalty (Section 3.17). Classification proceeds.

**Step 2: Tokenisation.** WordPiece yields a short sequence with `[CLS]`, symptom subwords, `[SEP]`, and pads to length $M = 128$. Only the first few positions carry clinical content; pads are masked in attention. The map $\tau$ is deterministic: the same string always produces the same IDs.

**Step 3: Embeddings and $H^{(0)}$.** Each of the $M$ positions receives a 768-dimensional vector

$$H^{(0)}_{i,:} = E_{\text{word}}(t_i) + E_{\text{pos}}(i) + E_{\text{seg}}(0).$$

Shape $H^{(0)} \in \mathbb{R}^{128\times 768}$: rows are tokens, columns are features (Section 3.10).

**Step 4: Twelve encoder layers.** Self-attention mixes information across rows. After layer $\ell = 12$, row 0 is the summary $h_{[\text{CLS}]} \in \mathbb{R}^{768}$. Schematic attention (Table 3.14) illustrates higher weight on symptom tokens than on fillers.

**Step 5: Softmax.** Using the illustrative logits of Section 3.13,

$$\hat{\mathbf{y}} \approx [0.03,\ 0.07,\ 0.69,\ 0.15,\ 0.05], \qquad \hat{c} = 3.$$

This is a Categorical distribution with mode Level 3.

**Step 6: Colour and pathway.**

$$C_{\text{NLP}} = f_{\text{SATS}}(3) = \text{Yellow}.$$

The pathway card is

$$P(\text{Yellow}) = \big(60\,\text{min},\ \text{Outpatient urgent stream},\ \text{actions},\ \text{escalation}\big)$$

as in Table 3.22. Nursing sets a 60-minute reassessment timer; Diagnostics may run in parallel.

**Step 7: Parallel TEWS (when vitals arrive).** Suppose later measurements give TEWS integer $T = 4$. Then $C_{\text{TEWS}}$ is Yellow on the SATS vital bands (Section 3.3). Text colour and vital colour agree in this illustration; if they disagreed, nurse judgment and SATS discriminators govern, not automatic Softmax override.
The walkthrough shows why every mathematical object in Chapter 3 appears in the deployed API response: gate metadata, $\hat{\mathbf{y}}$, $C_{\text{NLP}}$, and pathway fields.

## 3.19 End-to-end summary

For any chief complaint $X$, the implemented text path (matching the prototype `/predict` endpoint) proceeds as follows:

$$\begin{aligned} X &\xrightarrow[3.17]{g} \tau(X) \xrightarrow[3.7\text{--}3.10]{} H^{(0)} \\ &\xrightarrow[3.11\text{--}3.12]{12} h_{[\text{CLS}]} \\ &\xrightarrow[3.13]{} \hat{\mathbf{y}} \xrightarrow[3.15]{} C_{\text{NLP}} \xrightarrow[3.16]{} P(C_{\text{NLP}}), \end{aligned} \tag{3.33}$$

where each arrow label points to the section that defines that transformation: $g$ is the clinical gate, $\tau(X)$ is tokenisation, $H^{(0)}$ is the input embedding matrix, $h_{[\text{CLS}]}$ is the final summary vector, $\hat{\mathbf{y}}$ is the five-class Softmax distribution, $C_{\text{NLP}} = f_{\text{SATS}}(\hat{c})$ is the SATS colour, and $P(C_{\text{NLP}})$ is the pathway card. If $g(X) < 0.30$, the chain stops after the gate and no colour is assigned.

1. **Clinical relevance gate.** Compute $g(X)$; if $g(X) < 0.30$, return a request for clinical detail and stop.
2. **Tokenisation and embedding.** Map $X$ to tokens, build $H^{(0)} \in \mathbb{R}^{M\times768}$ (Sections 3.7–3.10).
3. **BioBERT encoding.** Apply $L = 12$ encoder layers with self-attention (Sections 3.11–3.12) to obtain $h_{[\text{CLS}]}$.
4. **Classification.** Apply the softmax head (Section 3.13):

$$\begin{aligned} \hat{\mathbf{y}} &= \text{softmax}(Wh_{[\text{CLS}]} + \mathbf{b}), \\ \hat{c} &= \arg\max_c \hat{y}_c. \end{aligned}$$

5. **Colour and pathway.** Apply $C_{\text{NLP}} = f_{\text{SATS}}(\hat{c})$ and return pathway card $P(C_{\text{NLP}})$ (Sections 3.15–3.16).

The illustrative complaint in this chapter instantiates the same pipeline; one worked outcome is level 3 (Yellow) at 72% confidence for headache and feverish symptoms.

TEWS (Section 3.3) remains the nurse-facing vital-sign score when measurements become available.

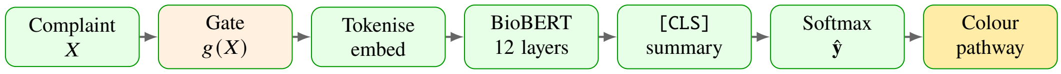


Figure 3.11: Compact end-to-end NLP triage path (TEWS documented separately in Section 3.3).

The diagram above matches the pipeline in Section 3.1. In order:

1. **Complaint** $X$ enters as free text from the patient or attendant.
2. **Gate** computes $g(X)$; non-clinical input is rejected before BioBERT runs.
3. **Tokenise and embed** map text to integer IDs and build the matrix $H^{(0)}$.
4. **BioBERT** applies twelve encoder layers with self-attention across all tokens.
5. **`[CLS]` summary** extracts the fixed-length vector $h_{[\mathrm{CLS}]}$ from row 0.
6. **Softmax** outputs acuity probabilities $\hat{\mathbf{y}}$ over five classes.
7. **Colour and pathway** apply $f_{\mathrm{SATS}}(\hat{c})$ and return the pathway card $P(C_{\mathrm{NLP}})$.

## 3.20 Implementation note

Figure 3.12 shows the deployable system: patient-facing web interface, chatbot API, and hospital routing layer. The NLP mathematics in Figures 3.1 and 3.11 execute inside the API; department routing applies facility pathway rules after classification.

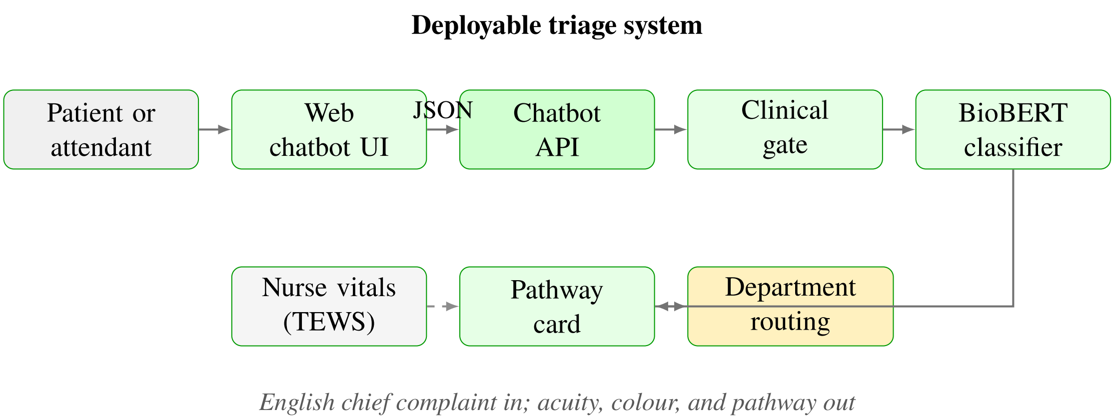


Figure 3.12: System architecture: user interface, chatbot API, BioBERT inference pipeline, pathway card, and department routing. Nurse TEWS runs in parallel (dashed); prototype code in Appendix A.

The prototype exposes a chatbot application programming interface (API) endpoint: JSON **English** chief complaint in; acuity probabilities, $C_{\mathrm{NLP}}$, pathway fields, and gate metadata out. Model weights and tokenizer files ship with the repository listed in Appendix A. This project treats the mathematics above as the specification; measured results appear in Chapter 4.

# CHAPTER 4

# RESULTS

Evaluating the BioBERT triage classifier requires a structured analysis of its foundational architecture, predictive performance under domain shift, and pathway implications. The initial phase establishes the computational baseline using the twelve-layer BioBERT architecture defined in Chapter 3. Primary reported metrics then come from **external** evaluation on the National Hospital Ambulatory Medical Care Survey (NHAMCS) 2018–22 and the Medical Information Mart for Intensive Care IV Emergency Department (MIMIC-IV-ED) open demo matching the v2.2 triage schema (National Center for Health Statistics (NCHS), 2024; Johnson et al., 2023b,a). Triagegeist is used for training and validation only. None of these sources is prospective live performance at KNUST University Hospital.

## 4.1 BioBERT model analysis

The first results question is not “how accurate is the chatbot on a table?” but “what model is deployed, and does it work only after the right training?” Before project-specific fine-tuning, the classifier rests on BioBERT-base (Lee et al., 2020): a twelve-layer bidirectional encoder with hidden dimension 768, twelve attention heads, project maximum sequence length $M = 128$ tokens (native BioBERT maximum is 512; Section 3.7), and a WordPiece vocabulary of approximately 28,996 tokens (Section 3.11). Stages 1 and 2 (Section 3.14) supply biomedical language understanding and partial triage-oriented encoder features; they do *not* produce the five-class acuity mapping required at intake. Stage 3 reinitialises the classification head and trains on the Triagegeist English corpus (Section 3.1.2). Class imbalance in the training corpus (Level 3 is the mode; Level 1 is the minority) motivates Stage 3 stratified oversampling documented in Section 4.2.

Figure 4.1 summarises the three-stage training stack. Stage 1 applies masked language modelling on PubMed and PubMed Central text; Stage 2 transfers encoder weights from a prior two-class triage checkpoint with the old head discarded; Stage 3 is the comprehensive project fine-tuning that maps free-text complaints to five ESI-style acuity levels.

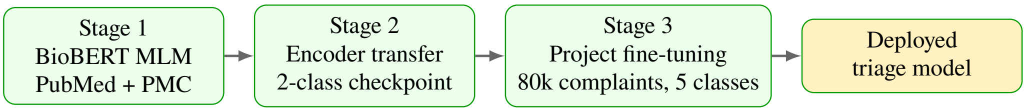


Figure 4.1: Three-stage BioBERT training stack (Stages 1–2 pre-date project fine-tuning; Stage 3 produces the deployed five-class head).

Read left to right, the diagram is a horizontal training pipeline: each box is one stage and the arrows show how parameters flow into the next. Stage 3 is the only stage that maps chief complaints to five acuity levels. The chatbot's urgency signal therefore depends on completing Stage 3; Stages 1–2 alone cannot route patients safely.
Naive bounds still motivate Stage 3: uniform random and an untrained head sit near 20% five-class accuracy, and a majority-class rule (always Level 3) reaches about 36% on Triagegeist with 0% Red recall. Stages 1–2 alone therefore cannot route patients. Published headline performance is deferred to the external evaluation in Section 4.2 (Table 4.2 and Figure 4.3).
Figure 4.2 visualises the static dimensions of the deployed encoder.

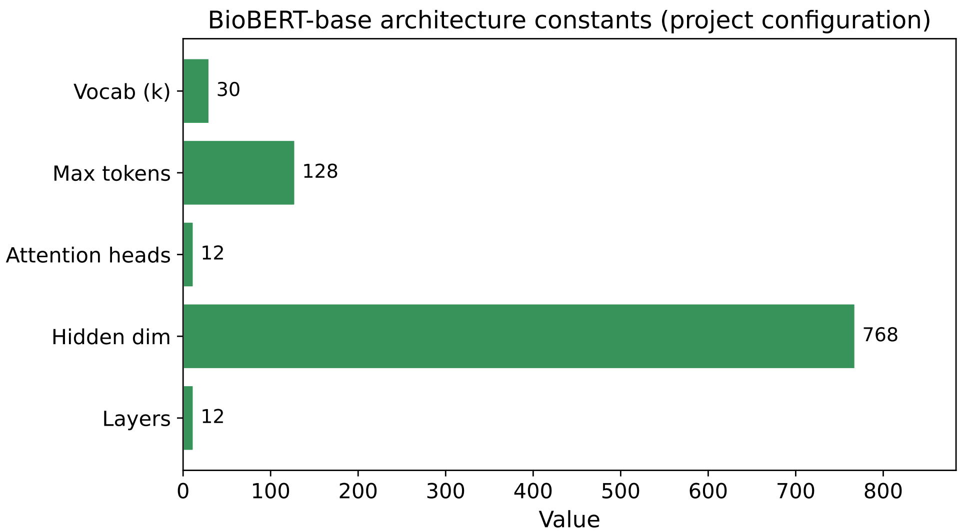


Figure 4.2: BioBERT-base architecture constants (project configuration; $M = 128$ is a deployment choice versus native length 512).

Figure 4.2 lists architecture parameter names on the vertical axis and their numeric values on the horizontal axis, drawn from the project `config.json` and the BioBERT-base specification (Lee et al., 2020). Layers (12), hidden dimension (768), and attention heads (12) are fixed by BioBERT-base. The maximum token length is deliberately constrained to $M = 128$ (versus the native BioBERT maximum of 512) to keep intake inference sub-second; the vocabulary is approximately 30,000 tokens.

## 4.2 Fine-tuned model analysis

Having established the BioBERT configuration and the necessity of Stage 3 fine-tuning, the remainder of this chapter reports **external** predictive metrics, colour accuracy via $f_{\text{SATS}}$, under/over-triage, and pathway implications.

### 4.2.1 Evaluation protocol

Primary reported metrics are from external evaluation on MIMIC-IV-ED and NHAMCS. Triagegeist is used for training/validation only. None of these sources is prospective live performance at KNUST University Hospital.

1. **Train / validation (Triagegeist).** Stratified 90/10 split (seed 42) for Stage 3 fine-tuning and checkpoint selection. No Triagegeist holdout accuracy is reported as a Results headline.

2. **External A: MIMIC-IV-ED.** Text-only on `chiefcomplaint` $\rightarrow$ predict acuity; gold = `acuity` $\in \{1, \ldots, 5\}$. Full MIMIC-IV-ED v2.2 is credentialed on PhysioNet (Johnson et al., 2023a); this project evaluates the open demo matching the same triage schema (Johnson et al., 2023b) ($N = 207$ after filters) until a full extract is available.

3. **External B: NHAMCS 2018–22.** Text-only on `chief_complaint_text` $\rightarrow$ predict acuity; gold = `target_triage_acuity` (National Center for Health Statistics (NCHS), 2024). After filters, $N = 58{,}124$ rows remain; metrics below use a stratified subsample of $N = 12{,}000$ (seed 42) for compute, with full-set class support reported in Table 4.1.

**Pre-processing (both external sets).** Drop empty, NaN, or whitespace-only complaints; keep acuity in $\{1, \ldots, 5\}$; do *not* expand nursing abbreviations (e.g. CP, SOB); tokenise with BioBERT WordPiece and truncate/pad to project length $M = 128$.
Two checkpoints are compared:

1. **Baseline fine-tuned BioBERT** (project default): end-to-end fine-tuning without minority rebalancing beyond the natural train distribution.

2. **Stratified-oversampling variant:** text augmentation plus random minority oversampling on the Triagegeist training split, then end-to-end fine-tuning (Section 4.2.4).

Metrics: five-class accuracy and macro-F1; 4-colour accuracy after $f_{\text{SATS}}$; under-triage (predicted less urgent than true); over-triage (more urgent); critical under-triage (true Red$\rightarrow$non-Red; true Orange$\rightarrow$Yellow/Green); latency.

### 4.2.2 Acuity distribution on test sets

Table 4.1 reports gold acuity counts by level after the shared pre-processing in Section 4.2.1. The NHAMCS counts come from the 2018–22 US emergency-department survey text field `chief_complaint_text` with gold labels `target_triage_acuity` (National Center for Health Statistics (NCHS), 2024): $N$ = 58,124 rows after filters, with the stratified evaluation subsample ($N$ = 12,000) used for published metrics. The MIMIC counts come from the open MIMIC-IV-ED demo on PhysioNet (Boston ED phrasing), mapping `chiefcomplaint` to gold `acuity` (Johnson et al., 2023b) ($N$ = 207 after filters). Real emergency departments are imbalanced: NHAMCS is dominated by Levels 3–4, and Level 1 is scarce (846 of 58,124; 1.5%). MIMIC demo counts are small, especially Levels 4–5, so rare-class metrics there are interpreted cautiously.

| **Set** | $N_1$ | $N_2$ | $N_3$ | $N_4$ | $N_5$ | **Total** |
|---|---|---|---|---|---|---|
| NHAMCS (full retained) | 846 | 8,597 | 29,568 | 16,715 | 2,398 | 58,124 |
| NHAMCS (eval subsample) | 175 | 1,775 | 6,104 | 3,451 | 495 | 12,000 |
| MIMIC-IV-ED demo | 18 | 97 | 90 | 2 | 0 | 207 |

Table 4.1: Gold acuity counts by level on the NHAMCS and MIMIC-IV-ED demo test sets after pre-processing.

### 4.2.3 Why prior in-distribution accuracy looked inflated

A Triagegeist duplicate analysis finds only 4,949 unique normalised complaint strings among 80,000 training rows (duplicate rate ≈ 93.8%; 4,184 templates appear at least five times). High template reuse helps explain why earlier in-distribution holdout figures near 99.9% looked unrealistically strong: they largely measure agreement on repeated synthetic phrasing, not generalisation. Those figures are therefore retired from the Results headline in favour of MIMIC/NHAMCS metrics below.

### 4.2.4 Stratified oversampling for class imbalance

Stage 3 training on Triagegeist is class-imbalanced: Level 3 (Urgent) is the mode, while Level 1 (Resuscitation / Red) is the minority. Rebalancing the training split can raise Red recall, but often at a cost to precision and overall accuracy. Two natural ways to reduce that imbalance were available for this project.

**A: Classical embedding SMOTE.** Raw chief-complaint strings are discrete, so SMOTE cannot interpolate between token sequences directly (Chawla et al., 2002). One workable route is to freeze (or otherwise detach) BioBERT, extract fixed feature vectors such as $h_{[\mathrm{CLS}]} \in \mathbb{R}^{768}$, apply SMOTE in that continuous space to synthesise minority examples, and train a *separate* linear classification head on the resulting embedding set. Gradients never update the text encoder during that second stage.

**B: Stratified oversampling (chosen).** Keep the end-to-end Stage 3 pipeline of Chapter 3: WordPiece tokenisation, BioBERT encoder, and Softmax head trained jointly. On the Triagegeist *training* split only, apply text augmentation to minority urgent classes, then random minority oversampling toward configured target counts, and continue full encoder-plus-head updates. The training configuration sets `USE_EMBEDDING_SMOTE=False`, so embedding-space SMOTE is not part of the published second checkpoint.

This project adopts **B** for the published imbalance-mitigated checkpoint and does *not* report a classical embedding-SMOTE model as the official second result.

The choice follows from the training objective. End-to-end Stage 3 is meant to adapt WordPiece and attention features to short chief-complaint language; that requires backpropagation into BioBERT. SMOTE-generated points in embedding space cannot be pushed back through the text encoder unless BioBERT is frozen, which would abandon that adaptation goal and replace it with a shallow head on static features. Stratified oversampling instead keeps real or lightly paraphrased complaint text inside the same tokenizer → encoder → Softmax path already specified in Chapter 3. Calling B "SMOTE" would therefore be inaccurate for the panel: Chawla-style SMOTE (Chawla et al., 2002) describes A, which was considered but not used for the published checkpoint. The NHAMCS and MIMIC comparisons below therefore contrast the baseline Stage 3 model with this stratified-oversampling checkpoint, not with an embedding-SMOTE head.

### 4.2.5 NHAMCS external results

On the stratified NHAMCS subsample ($N = 12{,}000$), baseline BioBERT reaches 47.60% five-class accuracy, 0.288 macro-F1, and 49.95% colour accuracy after $f_{\text{SATS}}$ (Table 4.2). Under-triage is 28.39% and over-triage 24.01%. Level 1 recall is only 8.00% (support 175) despite 82.35% Level 1 precision: most true Red cases are predicted as Orange or Yellow (Figure 4.5), which is the primary safety concern under domain shift (Berkowitz et al., 2019).

| Metric | NHAMCS base. | NHAMCS overs. | MIMIC base. | MIMIC overs. |
|---|---|---|---|---|
| $N$ | 12,000 | 12,000 | 207 | 207 |
| Accuracy | 0.4760 | 0.3045 | 0.3043 | 0.2126 |
| Macro-F1 | 0.2882 | 0.2146 | 0.1748 | 0.1781 |
| Colour accuracy | 0.4995 | 0.3422 | 0.3043 | 0.2126 |
| Under-triage | 0.2839 | 0.3903 | 0.6377 | 0.6087 |
| Over-triage | 0.2401 | 0.3052 | 0.0580 | 0.1787 |
| Recall L1 (Red) | 0.0800 | 0.2171 | 0.1111 | 0.5000 |
| Precision L1 | 0.8235 | 0.0597 | 0.4000 | 0.2647 |
| Latency p50 (ms) | 94.00 | 96.18 | n/a | n/a |

Table 4.2: External classification, colour, under/over-triage, and latency metrics. MIMIC columns use the open demo ($N$ = 207). MIMIC latency is reported as n/a: published timing uses the NHAMCS evaluation run (Section 4.2.8).

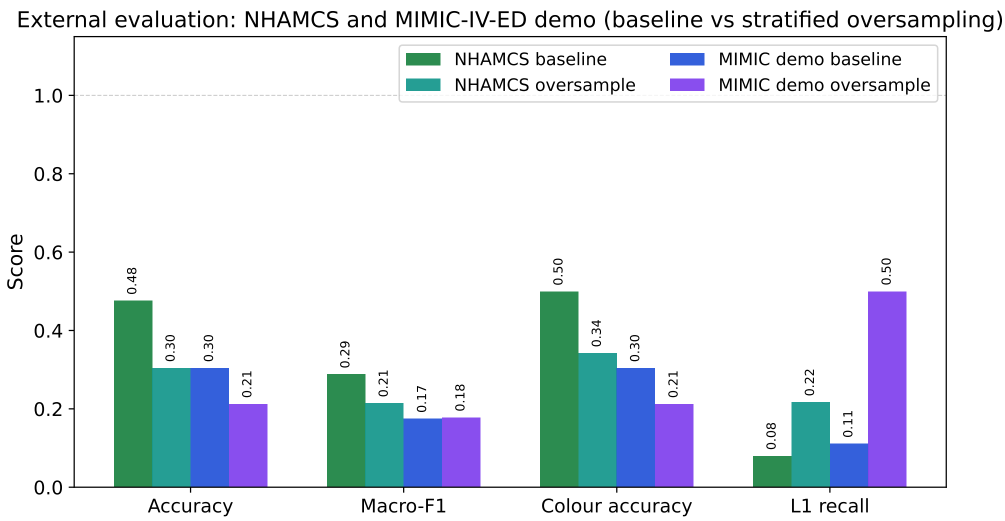


Figure 4.3: External headline metrics on NHAMCS ($N$ = 12,000 stratified subsample) and MIMIC-IV-ED demo ($N$ = 207): baseline versus stratified oversampling.

Figure 4.3 plots the same accuracy, macro-F1, colour accuracy, and Level 1 recall values as Table 4.2, so the baseline–oversample contrast is readable before the confusion matrices below.

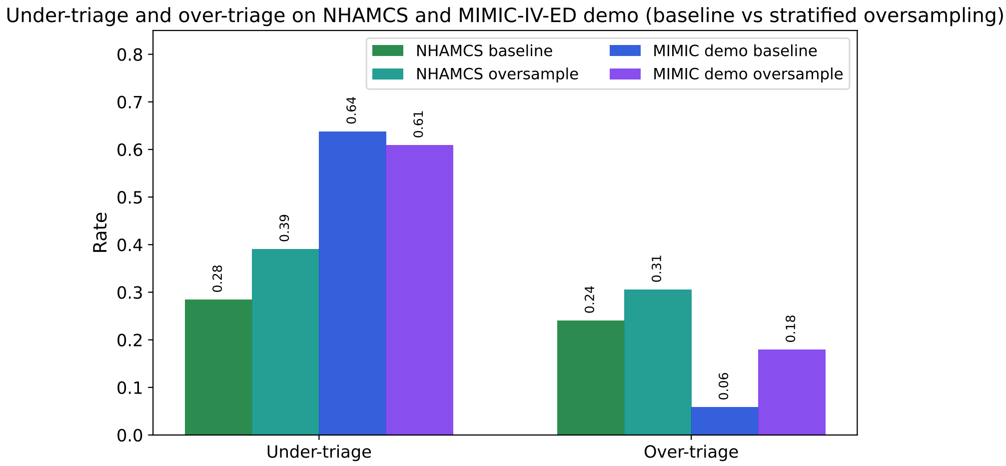


Figure 4.4: Under-triage and over-triage rates on NHAMCS ($N$ = 12,000 stratified subsample) and MIMIC-IV-ED demo ($N$ = 207): baseline versus stratified oversampling.

Under-triage (predicted less urgent than gold) is the safety failure mode: urgent language is routed too late. Over-triage (predicted more urgent than gold) is false escalation that loads acute streams and staff. Figure 4.4 shows that on NHAMCS, stratified oversampling raises *both* under-triage (28.39% to 39.03%) and over-triage (24.01% to 30.52%) even while Level 1 recall rises (Figure 4.3, Table 4.2), so sensitivity gains are not a free improvement in directional error. On the MIMIC demo, baseline under-triage is already high (63.77%); oversampling mainly increases over-triage (5.80% to 17.87%). These patterns reinforce mandatory nurse override under domain shift (Berkowitz et al., 2019).

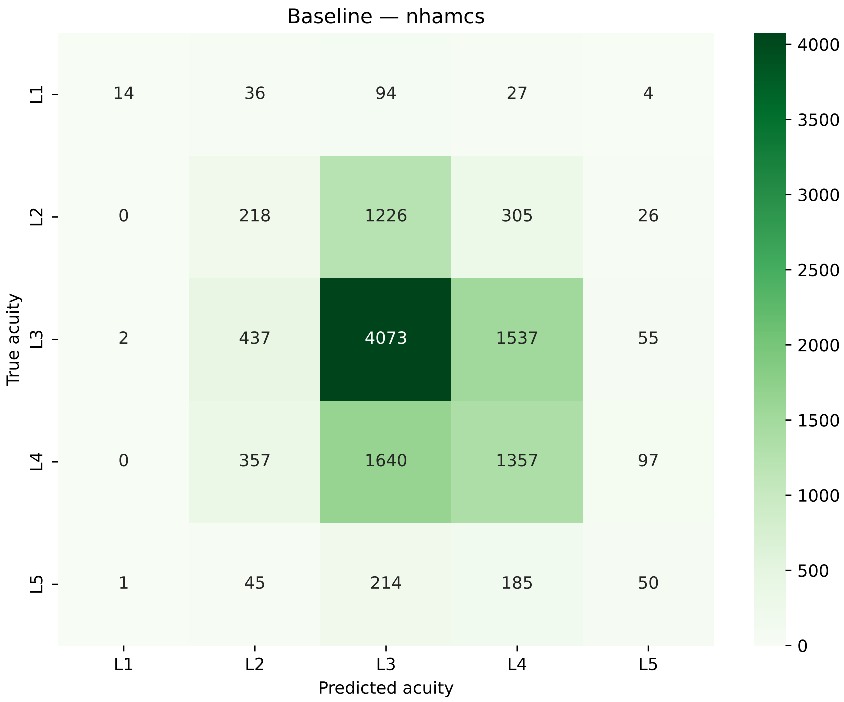


Figure 4.5: Baseline BioBERT confusion matrix on NHAMCS stratified subsample ($N$ = 12,000). Rows = true acuity; columns = prediction. Dark diagonal cells are correct predictions; off-diagonal mass is triage error.

The NHAMCS baseline matrix shows substantial off-diagonal mass. Of 175 true Level 1 cases, only 14 are recovered; 36 are predicted Level 2 (Orange) and 94 Level 3 (Yellow). Under-triage of critical presentations therefore dominates residual error on this US survey set.

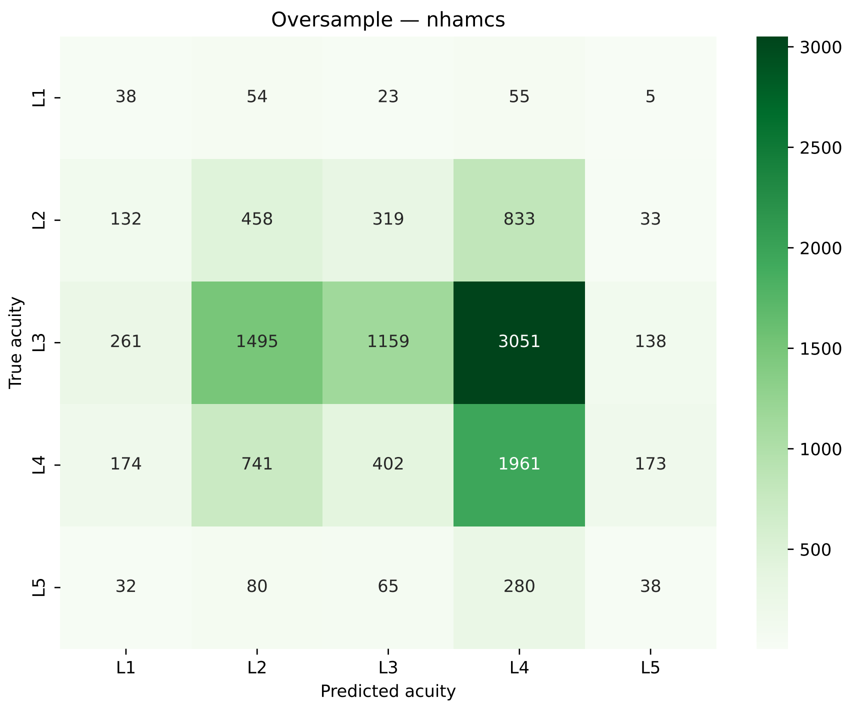


Figure 4.6: Stratified-oversampling BioBERT confusion matrix on the same NHAMCS subsample. Dark diagonal cells are correct predictions; off-diagonal mass is triage error.

Stratified oversampling raises NHAMCS Level 1 recall from 8.00% to 21.71% but lowers overall accuracy to 30.45% and Level 1 precision to 5.97%: more false urgent alerts accompany the sensitivity gain. As visually demonstrated by the drop in the Accuracy bars in Figure 4.3, stratified oversampling trades overall exact-match accuracy to raise Level 1 recall (Table 4.2). The baseline checkpoint remains the project default unless stakeholders explicitly prioritise Red sensitivity over precision (Berkowitz et al., 2019).

### 4.2.6 MIMIC-IV-ED demo results

On the open MIMIC-IV-ED demo ($N$ = 207), baseline accuracy is 30.43% with colour accuracy 30.43% and high under-triage (63.77%). As shown in the baseline matrix (Figure 4.7), the model correctly recovers only 2 of 18 true Level 1 cases. With stratified oversampling (Figure 4.8), this increases to 9 of 18 cases (50.00% recall), though the matrix shows a corresponding spread of false positives across Levels 2 and 3, and overall accuracy falls to 21.26%. These demo figures validate the evaluation pipeline on real Boston ED phrasing but are not substitutes for a full credentialed MIMIC-IV-ED v2.2 extract (Johnson et al., 2023a).

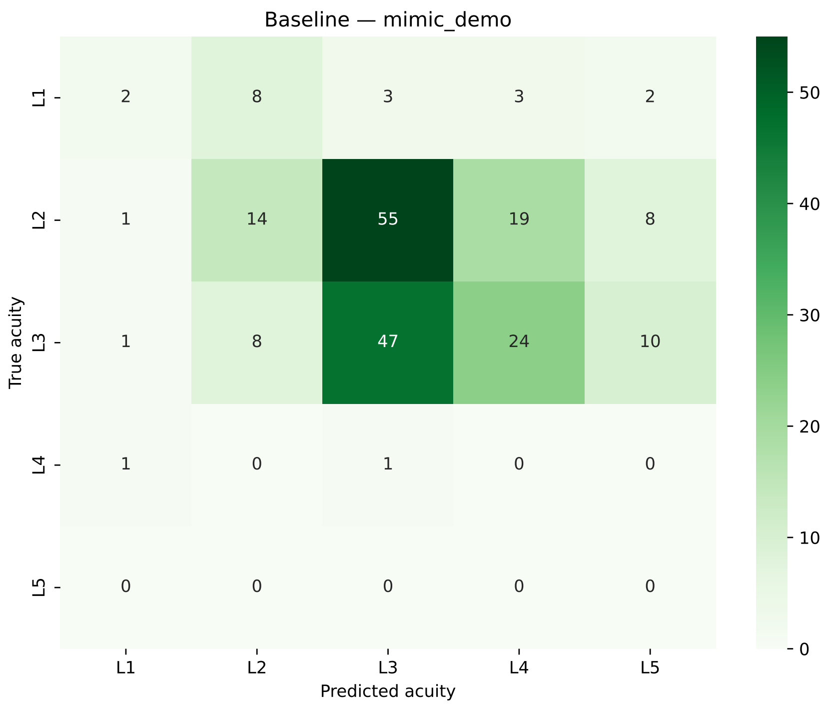


Figure 4.7: Baseline BioBERT confusion matrix on MIMIC-IV-ED demo ($N = 207$). Rows = true acuity; columns = prediction. Dark diagonal cells are correct predictions; off-diagonal mass is triage error.

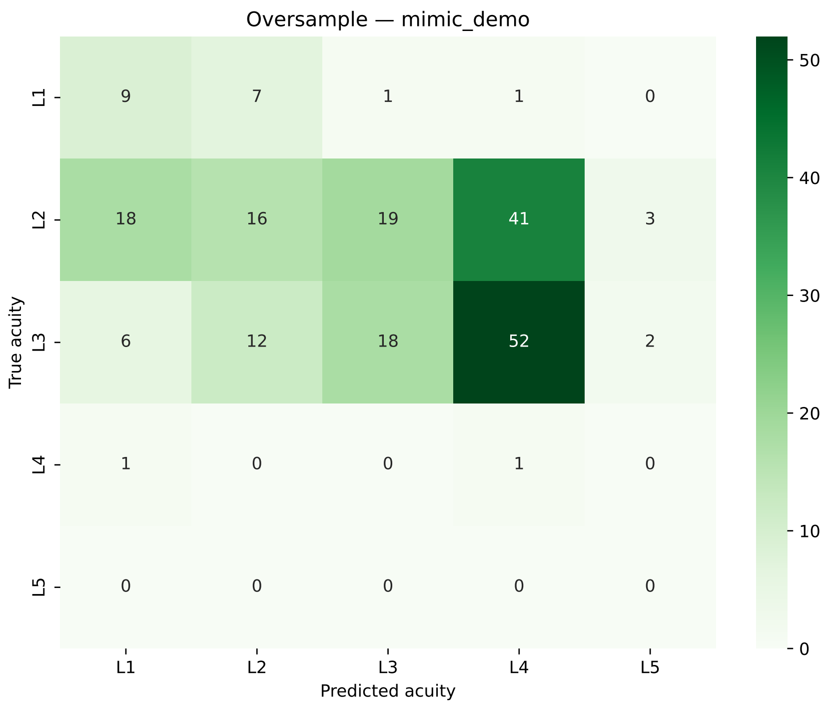


Figure 4.8: Stratified-oversampling BioBERT confusion matrix on MIMIC-IV-ED demo. Dark diagonal cells are correct predictions; off-diagonal mass is triage error.

### 4.2.7 Colour accuracy and under/over-triage

Mapping levels through $f_{\text{SATS}}$ (Table 3.20) yields 4-colour accuracy at or above five-class accuracy when Levels 4 and 5 both map to Green. On NHAMCS baseline, colour accuracy (49.95%) slightly exceeds five-class accuracy (47.60%). Critical under-triage of true Red to non-Red remains high (92.00% baseline NHAMCS; 78.29% oversampled), reinforcing mandatory nurse override for urgent language (Berkowitz et al., 2019).

### 4.2.8 Inference latency

Real-time intake requires sub-second response (Stewart et al., 2023). Latency in Table 4.2 is a property of the BioBERT forward pass at project length $M = 128$ on the evaluation hardware, not a dataset-specific accuracy score. The published timing benchmark is therefore the NHAMCS stratified subsample ($N = 12{,}000$), which is large enough for stable median and tail estimates: baseline median latency $\approx 94\,\text{ms}$ (p95 $\approx 124\,\text{ms}$); the stratified-oversampling checkpoint is similar (p50 $\approx 96\,\text{ms}$). Both respond in well under one second, supporting chatbot intake before nurse vitals are recorded.

MIMIC-IV-ED demo columns in the same table show n/a for latency by design. That demo ($N = 207$) is used for external accuracy, colour accuracy, and under/over-triage only. A second latency estimate on so few batches would not change the operational sub-second claim and would overstate the precision of a short timing sample. The same checkpoints and the same $M = 128$ truncation are used on both externals, so the NHAMCS p50 already represents chatbot response time at intake.

**Remark 4.1.** *External metrics are substantially lower than earlier Triagegeist in-distribution figures, as expected under geographic and demographic domain shift (Chapter 5). Text still carries a partial acuity signal, but the chatbot must remain nurse-overridable.*

## 4.3 Pathway routing at KNUST University Hospital

The complete five-scenario pathway map (gate reject plus Red, Orange, Yellow, Green) is specified in Tables 3.22 and 3.23 and illustrated in Figure 3.10. Each scenario assigns primary clinical departments, support actions across Nursing, Pharmacy, Health Information, Hospital Information Technology, Administration, and Accounts, and an escalation rule (South African Triage Group, 2012; Rominski et al., 2014).

### 4.3.1 Nurse triage vs chatbot response

Table 4.3 contrasts manual nurse triage with the hospital chatbot on dimensions relevant to intake at KNUST University Hospital. Nurse-side timings are drawn from published LMIC triage literature; chatbot timings are measured on the external evaluation hardware.

| Dimension | Manual nurse triage | Hospital chatbot (this project) |
|---|---|---|
| Time to first acuity signal | Queue-dependent; LMIC intake often delayed minutes to hours before nurse contact (KATH Emergency Department, 2018; Mitchell et al., 2023; Nurchis et al., 2025) | Approximately 94–100 ms inference on evaluation hardware (Table 4.2) |
| Inputs used | Chief complaint + vitals + TEWS + clinical judgment (South African Triage Group, 2012) | Text-only chief complaint; TEWS applied separately by nurse (Section 3.3) |
| Output | SATS colour + bedside assessment | SATS colour + pathway card + confidence score |
| Validation basis | Live clinical decision at triage desk | External offline agreement on MIMIC/NHAMCS text (Section 4.2.5) |
| Role at KNUST University Hospital | Authoritative triage | Triage assist at arrival; nurse override mandatory for low confidence and all vitals (Stewart et al., 2023) |

Table 4.3: Manual nurse triage versus chatbot assist at intake.

Once the chatbot assigns a colour in approximately 100 ms, pathway timer rules take effect: Orange cases require clinician contact within ten minutes, Yellow within 60 minutes, and Green within four hours (Section 3.16, Table 3.22) (South African Triage Group, 2012; Sundström et al., 2014). The chatbot does not replace nurse assessment; it provides an immediate acuity signal so that queue time before *any* routing decision is reduced relative to waiting for a free triage nurse (Mitchell et al., 2023; Nurchis et al., 2025).

**Gate rejection.** Non-clinical input receives no colour; Nursing re-prompts for a chief complaint. This prevents meaningless acuity scores and keeps acute queues free of idle interactions.

**Red.** Life-threatening text (e.g. crushing chest pain with dyspnoea) routes immediately to Surgery and Internal Medicine with Code Red alerts via HIT. Administration bypasses routine registration so resuscitation is not delayed (South African Triage Group, 2012).

**Orange.** Very urgent complaints (e.g. severe abdominal pain in pregnancy) activate Internal Medicine, Surgery, or Obstetrics and Gynaecology within ten minutes, with stat

Diagnostics and a head-nurse escalation if unclaimed.

**Yellow.** For the running illustrative complaint (*"I have a headache and feel feverish. My body aches and I feel weak."*), the model assigns level 3 at approximately 72% confidence. The pathway card routes to the Outpatient urgent stream and parallel Diagnostics under the adult ASTS default; Pediatrics is used only when the complaint contains explicit child markers. Nursing sets a 60-minute reassessment timer.

**Green.** Routine presentations (e.g. dental pain without fever) divert to Outpatient, Dental, Eye, or Public and Preventive Health within four hours, reducing congestion in acute waiting areas (Yilmaz, 2021; Sundström et al., 2014).

Classification output (colour + confidence + department list) is the machine-readable input to this routing layer. External evaluation (Section 4.2.5) bounds offline reliability under domain shift; live intake at KNUST University Hospital still requires nurse override.

## 4.4 Anticipated impact on healthcare delivery

The following analysis interprets how the chatbot can support faster, clearer hospital flow at KNUST University Hospital, drawing on external metrics where available and on published triage literature elsewhere. No prospective waiting-time trial has yet been conducted at KNUST University Hospital (Mitchell et al., 2023).

### 4.4.1 Functional chatbot and workflow improvement

We anticipate a functional chatbot that suggests urgency from a single chief complaint and supports hospital workflow by assigning SATS colour and department direction at intake (Stewart et al., 2023; Mitchell et al., 2023). External NHAMCS evaluation reports 47.60% five-class accuracy and 49.95% colour accuracy for the baseline checkpoint (Section 4.2.5), indicating that text carries a partial acuity signal under domain shift rather than near-perfect agreement. Median inference latency of approximately 94–100 ms on evaluation hardware supports rapid digital response at arrival (Table 4.3).

Figure 4.9 compares time-to-first-acuity-signal: traditional intake (top) waits for a nurse; chatbot-assisted intake (bottom) classifies text immediately and runs nurse vitals in parallel.

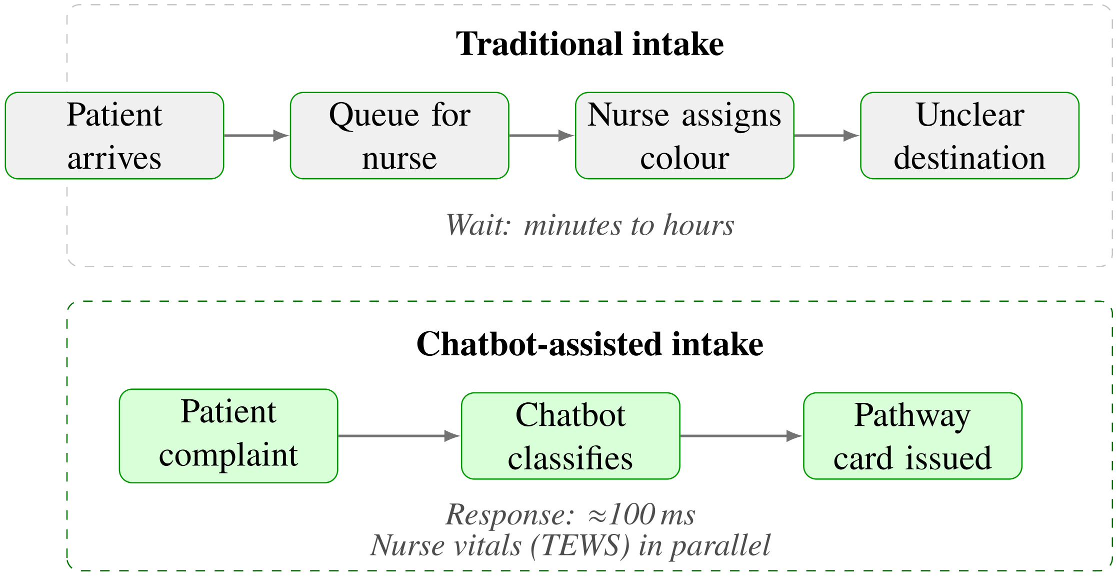


Figure 4.9: Stacked intake comparison at KNUST University Hospital. **Top:** patient queues minutes to hours before a nurse assigns colour. **Bottom:** chatbot classifies the complaint in ≈100 ms and issues a pathway card; nurse vitals continue in parallel. Colour routing (Red/Orange vs Green) is detailed in Section 4.3.

The stacked workflow diagram has two panels read left to right. The top panel is traditional intake (arrive → queue → nurse assigns colour → unclear destination); the bottom panel is chatbot-assisted intake (complaint → chatbot classifies → pathway card issued). Pathway design comes from Chapter 3, chatbot latency from the external evaluation in Section 4.2.8, and traditional wait times from LMIC triage literature (KATH Emergency Department, 2018; Mitchell et al., 2023). The top footnote reads "Wait: minutes to hours"; the bottom footnotes read "Response: ≈100 ms" and "Nurse vitals (TEWS) in parallel". Red/Orange fast-track and Green digital queue are developed in Section 4.3. The figure therefore contrasts slow nurse-queue intake with immediate chatbot classification while making clear that nurse assessment is not replaced (South African Triage Group, 2012; Stewart et al., 2023).

### 4.4.2 Reduced initial triage wait times

Published triage literature suggests wait times for initial triage may decrease when patients interact with the bot immediately rather than queue for a nurse (Mitchell et al., 2023; Nurchis et al., 2025). External evaluation shows the model can assign colour and pathway in approximately 100 ms once valid clinical text is submitted (Table 4.3). Non-urgent (Green) cases can be digitally queued for later consultation while urgent cases are fast-tracked through Red and Orange escalation rules (Table 3.22), potentially reducing overall waiting duration (Sundström et al., 2014; Yilmaz, 2021). Parallel diagnostics orders flagged on Yellow and Green pathway cards may further shorten total time to definitive care (Sundström et al., 2014).

### 4.4.3 Critical-case sensitivity

The chatbot algorithm, combined with Red/Orange pathway escalation, is designed to trigger alerts when language suggests life-threatening conditions (Gligorijević et al., 2018; South African Triage Group, 2012). On NHAMCS, baseline Level 1 recall is only 8.00% (support 175), with most missed Red cases predicted as Orange or Yellow (Figure 4.5). Stratified oversampling raises Level 1 recall to 21.71% at large cost to precision. High offline sensitivity for critical cases is therefore *not* demonstrated on US external text; nurse override and prospective audit remain essential (Berkowitz et al., 2019). Waiting-time deterioration increases adverse outcomes when triage colour is not reassessed (Nurchis et al., 2025); digital timers on Orange and Yellow pathway cards reduce this risk (South African Triage Group, 2012).

### 4.4.4 Patient satisfaction and appropriate categorisation

Patients may experience improved satisfaction when they receive quick feedback on urgency and feel guided through the system rather than left without direction (Stewart et al., 2023). Over time, the percentage of patients appropriately categorised (as confirmed by clinicians after assessment) will measure chatbot accuracy in live use. External offline agreement is a necessary but not sufficient requirement for clinical confirmation rates (Section 4.2.5).

### 4.4.5 Staff workload and public-health insights

Nursing staff may report reduced triage documentation burden, devoting more time to direct patient care and vitals (TEWS) rather than initial information gathering (Stewart et al., 2023; Mitchell et al., 2023). The gate rejects non-clinical messages before BioBERT, filtering idle interactions (Section 3.17). If scaled, anonymised complaint records could reveal public-health trends (e.g. common presentations by season); privacy and governance review would be required before aggregation.

### 4.4.6 Success criteria

Project success would be measured by smoother patient flow, colour-coded pathways guiding patients to the right place at the right time, and a reduction in adverse events or delays attributable to triage (Rominski et al., 2014; KATH Emergency Department, 2018). Published NLP triage systems elsewhere report streamlined processes and optimised resource use in busy hospitals (Stewart et al., 2023; Mitchell et al., 2023).

**Limitations.** External metrics measure classification accuracy on US chief-complaint text, not patient outcomes, waiting-time reduction, or satisfaction scores at KNUST University Hospital (Mitchell et al., 2023; Nurchis et al., 2025). Prospective local validation is required. Training-data class imbalance and NHAMCS Level 1 scarcity widen uncertainty on rare urgent classes; deployment should treat low-confidence or contradictory presentations as mandatory nurse review (Berkowitz et al., 2019; South African Triage Group, 2012). Stratified oversampling increases Red recall on NHAMCS but sharply lowers Red precision; the baseline checkpoint is the project default unless clinical stakeholders prioritise sensitivity over precision.

## 4.5 Discussion of findings

Combining the model architecture analysis with the external metrics suggests three core conclusions, addressing the research objectives established in Chapter 1.

First, Stage 3 fine-tuning is required: naive bounds (uniform random near 20%; majority-class about 36% on Triagegeist with 0% Red recall) show that biomedical pre-training and encoder transfer alone are near chance for five-class acuity prediction. The Softmax head becomes useful for routing only after supervised fine-tuning on labeled chief complaints. Published performance under domain shift is then read from Figure 4.3 and Table 4.2, not from retired Triagegeist holdout headlines.

Second, the deterministic $f_{\text{SATS}}$ mapping ensures that colour routing errors are inherited from the acuity prediction $\hat{\mathbf{y}}$. Softmax probabilities therefore remain critical: a dominant Yellow prediction dictates a different operational response than a near-tie between Yellow and Green, reinforcing human oversight for low-confidence outputs.

Finally, clinical pathways operationalise the Softmax output. Tables 3.22–3.23 show how millisecond-scale text classification binds to hospital actions: timers, destinations, and support departments. Prospective time-motion data are still required for impact measurement; the architecture establishes an auditable protocol for intake assist, not unsupervised discharge.

**Safety posture.** External accuracy under domain shift does not license unsupervised discharge decisions. The deployed posture is triage assist with nurse override, TEWS in parallel, gate rejection of non-clinical text, and mandatory review for low Softmax confidence or text–vitals conflict (Stewart et al., 2023; Berkowitz et al., 2019). That posture is consistent with SATS training manuals and with under-triage quality literature.

# CHAPTER 5

# CONCLUSION

This project developed a mathematical pipeline from unstructured chief complaint text to South African Triage Scale (SATS) colour and clinical pathway at hospital intake. The chatbot accepts a single English free-text submission and returns acuity probabilities, a colour, and a pathway card aligned with SATS and healthcare-facility department structure.

1. A BioBERT classifier predicts ESI-style urgency from one English chief complaint. Primary reported metrics come from external evaluation on MIMIC-IV-ED and NHAMCS; Triagegeist is used for training and validation only.

2. The chatbot returns a SATS colour and a pathway card from text at intake, before vitals. Colour follows from the fixed map $f_{\text{SATS}}$; pathway cards $P(C_{\text{NLP}})$ name destination, $T_{\max}$, actions, and escalation, while TEWS remains the nurse vital-sign path in parallel.

3. A clinical gate with threshold 0.30 gives no colour to non-clinical chat. Non-medical messages are rejected before BioBERT runs.

Ultimately, this project demonstrates that advanced natural language processing can be safely mathematically constrained and operationalized for low-resource hospital settings. By bridging the gap between unstructured patient narratives and the deterministic SATS pathway protocols of KNUST University Hospital, this Flow Management Engine provides a rigorously auditable foundation for reducing triage bottlenecks and ensuring critical-case visibility from the moment a patient arrives.

## 5.1 Limitations and future work

**Data scope.** Triagegeist is used for Stage 3 training and validation only. Published classification metrics in Chapter 4 are computed on external MIMIC-IV-ED and NHAMCS text, after dropping empty complaints and retaining acuity labels in $\{1, \ldots, 5\}$, with tokenisation truncated to project length $M = 128$. The model is English-only at intake. An observational site visit to KNUST University Hospital mapped pathways and department structure through intake observation and staff inquiries; supplementary English example complaints generated from those protocols support prototype and qualitative analysis only and were not used for fine-tuning. A natural next step is prospective collection of nurse-labeled chief complaints at KNUST University Hospital for local retraining, and exploring optional fusion of vitals and TEWS alongside free text when measurements become available.

**Geographic and demographic domain shift.** The chatbot is designed for Ghanaian facility pathways at KNUST University Hospital, yet the external test corpora are United States sources. MIMIC-IV-ED is drawn from a Boston-area teaching hospital population that is predominantly White (approximately 68%), with smaller shares of Black (approximately 15%), Hispanic/Latino (approximately 5%), and Asian (approximately 3%) patients in published MIMIC demographic descriptions. NHAMCS likewise reflects a North American ambulatory and emergency survey population, not Ghanaian intake language mix, care pathways, or case-mix. External accuracy therefore does *not* equal expected live performance at KNUST University Hospital; prospective local validation remains essential.

**Class imbalance and Level 1 scarcity.** Real emergency departments are heavily skewed toward mid-acuity visits. NHAMCS in particular has very few Level 1 (Resuscitation) cases relative to Levels 3–4. Chapter 4 reports class support $N_\ell$ alongside recall so rare-class metrics are interpreted cautiously. Stratified oversampling on the Triagegeist training split studies the recall–precision trade-off; classical embedding-space SMOTE was not used for the published oversampled checkpoint (Section 4.2.4).

**Age and pediatrics.** The classifier has no age feature and defaults to adult SATS (ASTS) TEWS and pathway tables. Pediatrics is not a default Yellow destination; routing to Pediatrics is reserved for complaints with explicit child markers. Structured age input is future work.

**Clinical validation.** External offline evaluation suggests text carries a useful acuity signal under domain shift, but Softmax confidence should still trigger nurse oversight when the mode is not dominant or when language conflicts with vitals. The clinical relevance gate was validated only on a small hand-labeled contrast set ($N = 52$), not on prospective KNUST University Hospital traffic; lexicon-poor clinical wording can still produce false clinical rejections and must be handled by re-phrasing or nurse override (Section 3.17). The next phase is a prospective pilot at KNUST University Hospital to measure waiting times, nurse override rates, and patient outcomes in live intake (Mitchell et al., 2023).

**Pathway refinement.** The five-scenario pathway map is SATS-aligned and complete for gate reject, Red, Orange, Yellow, and Green routing, including Green diversion from acute queues. Ongoing collaboration with hospital administrators and clinical leads at KNUST University Hospital will keep department routing tables current as service lines evolve (Rominski et al., 2014; South African Triage Group, 2012).

**Language and equity.** The current model processes formal English. Expanding the system to handle Ghanaian informal English (Pidgin) or integrating voice-to-text capabilities for Twi would significantly broaden accessibility for patients who are less comfortable writing formal English complaints.

**System integration.** The prototype chatbot API and pathway cards already anchor the chatbot in a deployable pipeline. Deeper integration with Hospital Information Technology and Health Information systems would automate timer alerts and bed assignment, completing the Flow Management Engine vision across Administration, Diagnostics, and all clinical departments listed in Table 3.24.

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# Appendix A

# Software repository

The implemented prototype and fine-tuned BioBERT artefacts are maintained at:
`https://github.com/samuel6814/triage`
No full source code is reproduced in this project; the mathematical specification in Chapter 3 and the system architecture in Figure 3.12 are the primary references.

# Appendix B

# Prototype `/predict` response fields

A successful clinical submission returns a JSON object whose fields mirror the mathematics of Chapter 3. Table B.1 lists the principal keys.

| **Field** | **Meaning** |
|---|---|
| `gate_score` | Clinical relevance $g(X)$ in $[0, 1]$ |
| `gate_passed` | True if $g(X) \geq 0.30$ |
| `probabilities` | Softmax vector $\hat{\mathbf{y}}$ over levels 1–5 |
| `predicted_level` | $\hat{c} = \arg\max_c \hat{y}_c$ |
| `sats_colour` | $C_{\text{NLP}} = f_{\text{SATS}}(\hat{c})$ |
| `pathway` | Pathway card $P(C)$: $T_{\max}$, destination, actions, escalation |
| `confidence` | Mode probability $\hat{y}_{\hat{c}}$ for nurse display |

Table B.1: Principal `/predict` response fields and their mathematical counterparts.

Gate failures omit colour and pathway fields and instead return a prompt requesting a clinical chief complaint.

# Appendix C

# Stage 3 fine-tuning checklist

For reproducibility, Stage 3 training used the following checklist (full values in Table 3.18):

1. Join Triagegeist `train.csv` and `chief_complaints.csv` on `patient_id`.
2. Keep only the English free-text chief complaint (why the patient came) and the nurse-assigned urgency level (1–5); drop vitals and demographics.
3. Stratified 90/10 split with seed 42 ($N_{\text{train}} \approx 72{,}000$, $N_{\text{eval}} = 8{,}000$).
4. Load BioBERT encoder; discard any prior two-class head; initialise $W \in \mathbb{R}^{5\times768}$, $\mathbf{b} \in \mathbb{R}^{5}$.
5. Train three epochs, batch size 16, learning rate $2 \times 10^{-5}$, weight decay 0.01, max length 128, FP16 enabled.
6. Select the checkpoint with best validation cross-entropy (0.001812 at epoch 3 for the baseline run).
7. Optional stratified-oversampling branch: augment and randomly oversample minority urgent classes on the training split before the same training recipe; compare on external MIMIC/NHAMCS sets (not a Triagegeist holdout headline).

# Appendix D

# Glossary of principal symbols

| Symbol | Definition |
|---|---|
| $X$ | English chief-complaint string |
| $\tau$ | WordPiece tokenisation map |
| $M$ | Sequence length (maximum 128) |
| $D$ | Hidden dimension (768) |
| $H^{(\ell)}$ | Hidden-state matrix after encoder layer $\ell$ |
| $h_{[\mathrm{CLS}]}$ | Summary vector (row 0 of $H^{(12)}$) |
| $W, \mathbf{b}$ | Classification weight matrix and bias |
| $\mathbf{z}, \hat{\mathbf{y}}$ | Logits and Softmax (Categorical) probabilities |
| $\hat{c}$ | Predicted acuity level |
| $f_{\mathrm{SATS}}$ | Level-to-colour map |
| $C_{\mathrm{NLP}}$ | NLP-assigned SATS colour |
| $P(C)$ | Pathway card for colour $C$ |
| $g(X)$ | Clinical relevance gate score |
| $T$ | TEWS integer from vitals |

Table D.1: Compact glossary of symbols used across Chapters 3–4.